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Neuroevolution-based multiobjective algorithm for feature selection and binary classification of DNA microarrays

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ABSTRACT

The genomes of organisms have been sequenced for many years, leading to the discovery of thousands of genes. DNA microarrays are widely used tools for simultaneously analysing numerous genes, commonly employed in detecting and identifying various diseases, including cancer. However, microarray datasets have non-relevant and redundant information, hindering their analysis. This problem is further exacerbated considering the datasets' high dimensionality and imbalanced classes. Consequently, standard practice involves incorporating a feature selection process to identify the most relevant genes and their associations with diseases. Various methods have been employed to address this task. However, none have taken a more holistic approach that effectively handles feature selection, automatically identifies the optimal classifier configuration, and manages potential conflicting objectives simultaneously. In response, this study introduces the $\mathscr{S}$-metric selection - multiobjective neuroevolution of augmenting topologies (SMS-MONEAT) algorithm, which combines the multiobjective optimisation framework from $\mathscr{S}$-metric selection - evolutionary multiobjective algorithm (SMS-EMOA) and the evolutionary operators from the neuroevolution algorithm N3O, a variation from NEAT which stands for '3 new operators'. SMS-MONEAT algorithm was designed to perform both feature selection and optimise the configuration of artificial neural networks for classification tasks. SMS-MONEAT was evaluated against classic and state-of-the-art methods for feature selection and microarray classification. The experiments were conducted on 20 highly challenging cancer-type datasets primarily sourced from the Curated Microarray Database, and the results were investigated for statistical significance. The findings suggest that SMS-MONEAT either outperforms or achieves competitive results in terms of classification compared to the mentioned methods, while at the same time, it selects a smaller subset of features.

1. Introduction

Deoxyribonucleic acid (DNA) microarrays are a tool that allows the expression level analysis of thousands of genes simultaneously. A common application of this technology is detecting and diagnosing genetic diseases, such as cancer. Many efforts have been made to train predictive models for classifying microarray samples. However, analysing the information contained within microarrays is a challenging task. One of the reasons is the high dimensionality of microarray datasets since the number of samples is too small when compared with the number of features of each sample. Furthermore, microarray datasets tend to have imbalanced data, which means that the dataset has many more samples from one class than the others. These properties, inherent in these datasets, can cause conventional machine learning models to not generalise well to new samples. To approach this problem, it is recommended to include a feature reduction method [1].

Feature reduction is a procedure in which the number of variables from a dataset is decreased while keeping relevant information from each sample. The purpose is to remove redundant or non-relevant information from a dataset, making it easier and faster to analyse. Generally speaking, we can say that there are two main types of feature reduction: feature extraction and feature selection [2]. In the first one, new features are extracted, representing the relevant information with fewer features. The second method tries to find the subset of features that represent the relevant information. For microarray analysis, the latter is preferred because each feature is linked to a particular gene, and keeping the original feature representation could allow us to find relationships between specific genes and diseases [3].

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One popular method for feature selection is wrappers, which aim to find the subset of features that optimises the classification performance of a defined predictive model in an iterative process [1]. Several approaches have been proposed using evolutionary algorithms and swarm optimisation methods [4,5]. Moreover, multiobjective optimisation methods have been proposed to consider the number of features as an optimisation objective [6]. However, wrapper methods rely on the classification model used within it. Selecting which classification model to use and setting their parameters can be an enormous manual burden.

To assess this issue, neuroevolution approaches have been proposed for microarray analysis, performing the feature selection process and optimising the classification model simultaneously [7,8]. Neuroevolution optimises artificial neural networks (ANNs) using evolutionary computation methods.

The primary objective of this paper is to present a novel feature selection method for DNA microarray analysis. We introduce an innovative neuroevolution-based multiobjective algorithm, denoted as $\mathscr{S}$-metric selection - multiobjective neuroevolution of augmenting topologies (SMS-MONEAT). The algorithm was employed to maximise the predictive performance of ANN models while minimising the subset of selected features of microarray datasets.

The main contributions of this work are as follows,

1. A neuroevolution-based multiobjective algorithm, referred to as SMS-MONEAT, that simultaneously performs feature selection and optimises ANNs' topologies and their connection weight values.
2. A feature selection speciation method based on an external archive to enhance population diversity of SMS-MONEAT. This method splits the population according to the subset of selected features.
3. A feature selection and microarray classification pipeline which consists of three steps:
   (a) A feature selection filter based on the Kruskal Wallis H-test [9].
   (b) A feature selection wrapper based on the SMS-MONEAT algorithm that additionally generates a set of ANNs for microarray classification.
   (c) A decision-making step to select one individual from the output population using a weighted sum.
4. A hyperparameter tuning procedure for SMS-MONEAT and five state-of-the-art feature selection algorithms, conducted on microarray datasets using iRace [10].

The design of SMS-MONEAT was motivated by the microarray classification task. Nevertheless, SMS-MONEAT is a general approach for feature selection and optimising ANNs.

The rest of the paper is organised as follows. Section 2 explores related approaches that have inspired this study, such as feature selection approaches or multiobjective algorithms that have included NEAT concepts within their design. Section 3 provides an overview of the background concepts required for the development of this work. The proposed approach is detailed in Section 4. Section 5 outlines the experimental setup, and their results are discussed in Section 6. Finally, the conclusions of this research are given in Section 7.

2. Related works

This section describes different approaches proposed for the feature selection and microarray classification task. Additionally, it outlines some multiobjective algorithms that have incorporated NEAT elements in their design.

This paper represents an extension of the work presented at the Genetic and Evolutionary Computation Conference (GECCO) 2022 [11,12]. Both the methodology and experiments were refined to achieve more conclusive results.

2.1. Feature selection for microarray datasets

Feature selection is an optimisation task that removes non-relevant and redundant features from a particular dataset to improve the performance of predictive models. There are five types of methods for feature selection: filters, wrappers, embedded, hybrid and ensemble [1].

Filter methods use a statistical metric to evaluate the relationship between the class label and each feature. Several filter methods have been proposed for microarray feature selection [13,14], such as the Mann–Whitney U test, t test, Kruskal Wallis H test and Fisher F test.

Wrapper methods employ a searching strategy to select subsets of features and evaluate them according to the performance of a classification model using each subset. Evolutionary algorithms and swarm optimisation algorithms are popular techniques used as searching strategies in wrapper methods [4,5]. In particular, several wrapper methods have been proposed for feature selection in microarray datasets. These includes approaches based on techniques such as genetic algorithms [4,15–21], particle swarm optimisation (PSO) [22–26], ant colony optimisation (ACO) [27,28], artificial bee colony (ABC) [5,29], among others [30–32].

Embedded methods select features as part of the learning stage of the classification model [1]. Embedded methods based on algorithms such as recursive feature elimination [33,34] and randomised logistic regression [33] have been proposed for feature selection in microarray data.

Ensemble methods combine multiple subsets of features obtained either by using multiple feature selection methods with the same dataset (heterogeneous ensemble) or by using multiple datasets and the same feature selection (homogeneous ensemble) [1]. Multiple ensemble methods have been proposed for feature selection in microarray datasets [35–38].

Hybrid methods refer to approaches that combine two or more feature selection methods sequentially [39]. For example, some approaches compute a filter method previously to a wrapper method to reduce the search space and enhance the wrapper performance [4,8,17,38].

Approaches based on neuroevolution for feature selection in microarray data have been proposed in recent years. Grisci et al. [7] proposed a neuroevolution-based method for feature selection and microarray classification using a NEAT variant named Feature-Selective NEAT (FS-NEAT) [40]. Later, he presented a modified version of FS-NEAT, referred to as N3O [8], which stands for '3 new operators' and it included a modified crossover and two new mutation operators, which enhanced the feature selection exploration. They introduced N3O within a hybrid method for feature selection in microarray datasets, which consists of a filter method based on the Kruskal Wallis H test (KW) followed by a wrapper method based on N3O. One advantage of these neuroevolution-based methods is that they simultaneously perform the feature selection while optimising the ANNs used for classification.

The methods discussed thus far focused on optimising a single objective, which typically evaluates the classification model performance. However, the performance of a predictive model could be assessed in different ways. Rangasamy et al. [41] proposed using the multiobjective genetic algorithm (MoGA) as a wrapper for microarray feature selection using the classification model's accuracy, F-Value and geometric mean as optimisation objectives. Furthermore, several approaches have included the number of the selected features as an additional optimisation objective [6,42–46]. Dussaut et al. [6] specifically employed the maximisation of the classification model accuracy and the minimisation of the number of selected features as optimisation objectives and compared the performance of multiple algorithms such as Non-dominated Sorting Genetic Algorithm II (NSGA-II), Strength Pareto Evolutionary Algorithm 2 (SPEA2), Cellular Genetic Algorithm for Multiobjective Optimisation (MoCell), Multiobjective Cross-generational elitist selection,

Heterogeneous recombination, and Cataclysmic mutation (MOCHC), with MOCHC demonstrating superior overall performance.

The aforementioned multiobjective approaches used for feature selection in microarray datasets have the advantage of incorporating the number of selected features as an optimisation objective. Nevertheless, like most wrapper methods, they rely on the classification model. None of them include a method to optimise the classification model automatically, such as the one described in neuroevolution approaches.

Table 1 shows different feature selection approaches applied to microarray datasets. Three main aspects are highlighted: the types of feature selection used (Columns 2–5), whether the method was based on neuroevolution (Column 6), and whether it involved multiobjective optimisation (Column 7). Many of these approaches are hybrid, including the proposed method (last row). There is a tendency to use filter methods to reduce the search space before executing a wrapper, embedded, or ensemble method, which are more computationally expensive. Nevertheless, all of these approaches rely on different algorithms. Among all these approaches, only three are neuroevolution-based, which use FS-NEAT [7], N3O [8], and SMS-MONEAT (proposed algorithm) as wrapper methods. The advantage of neuroevolution approaches is that ANNs are optimised and perform the feature selection, reducing the manual burden of designing the ANN architecture. Furthermore, the great majority of the approaches presented are designed for single-objective optimisation, while using a multiobjective optimisation has the advantage of utilising multiple metrics to evaluate the classification model and including the number of features as an optimisation objective.

To the best of our knowledge, the proposed approach is the only one capable of performing feature selection while simultaneously optimising the topology and weights of ANNs, all while considering multiple objectives during the optimisation process.

2.2. NEAT for multiobjective optimisation

NEAT [49] is a neuroevolution algorithm whose purpose is to optimise the performance of ANNs on a specific task by evolving their topology and connection weights. The original NEAT design is for mono-objective optimisation. Nevertheless, many multiobjective algorithms have incorporated NEAT concepts in their designs [50–53].

Künzel et al. [53] introduced two evolutionary multiobjective algorithms based on NEAT: multiobjective NEAT (mNEAT) and mNEAT indicator-based (mNEAT-IB). Both algorithms employed quality indicators, such as R2 and the hypervolume, to evaluate the individual within the population. The main difference between mNEAT and the original NEAT is that mNEAT computes the R2 contribution of each individual and assigns it as their fitness; later, during the speciation process, each species is assigned a shared value based on their R2 contribution. By using the R2 contribution, the algorithm transforms the multiobjective optimisation problem into a mono-objective problem and executes NEAT as usual. On the other hand, mNEAT-IB combines the $\mathscr{S}$-metric selection - evolutionary multiobjective algorithm (SMS-EMOA) framework with NEAT evolutionary operators. SMS-EMOA is an indicator-based multiobjective algorithm which reduces the population by finding the individual with the lowest hypervolume contribution. The algorithm could be used either the hypervolume or R2 as the quality indicator metric. One modification from the original SMS-EMOA framework is that in mNEAT-IB, the offspring population can be larger than one, while in the original framework, it is equal to one.

Some NEAT variations such as FS-NEAT and N3O include techniques for performing feature selection within its execution, and they have been applied for feature selection and microarray classification tasks [7,8]. However, none of those variations have been incorporated within a multiobjective optimisation framework, which would allow them to include the feature reduction within the optimisation process.

3. Background

In this section, the feature selection problem is introduced along with a description of the methods employed in this study. Moreover, an introduction to multiobjective optimisation and neuroevolution is outlined.

3.1. Feature selection

A brief description of the filter and the wrapper method is provided in this subsection which were the methods employed in this work, however, as mention in Section 4.3, there are many more types of feature selection methods. Further information about them can be found in Alhewani et al. [1] study.

3.1.1. Filter methods

Filter methods employ a function to evaluate each feature independently and select features based on a defined threshold. It can be described as:

$$Q \subset M \ : \tag{1a}$$

$$f(X_q) < \delta, \tag{1b}$$

$$\forall q \in Q, \tag{1c}$$

where M is the full set of features and Q a subset from it; f is the filter function that gets as input a vector X_q, which stores the values from feature q. $f(X_q)$ output is compared against the threshold $\delta \in \{x \mid x \in \mathbb{R}, 0 \leq x \leq 1\}$ to determine whether the feature q is relevant.

3.1.2. Wrapper methods

Wrapper methods for feature selection are iterative methods that test the relevance of a subset of features in each iteration by training a predictive model and evaluating its performance. These methods use a search strategy to look for the subset of features that optimise the predictive model's performance. This work focuses mainly on predicted models based on supervised learning, which are machine learning models trained on tagged data. The optimisation problem can be described as:

$$\min_{X_Q \in X_M} \{g(y, y'_Q)\}, \tag{2a}$$

$$y'_Q = f(X_Q), \tag{2b}$$

$$Q \subset M, \tag{2c}$$

where M is the full set of features and Q a subset of it, thus, X_M represents the matrix of values from the full dataset and X_Q the dataset with just Q features; f is a predictive model which receives as input X_Q and outputs the predicted values y'_Q. The goal is to optimise the output from $g(y, y'_Q)$, which is a function to compare the predicted values (y'_Q) from the tagged values (y). In this example, $g(y, y'_Q)$ is visualised as the error between y_Q and y; therefore, the aim is to minimise it. Nevertheless, it could be aimed to maximise if a metric as the accuracy is used.

3.2. Multiobjective optimisation

Multiobjective optimisation methods come into play when two or more conflicting objectives must be simultaneously optimised. These techniques aim to find a set of solutions that balance the trade-off between the objectives, known as Pareto-optimal solutions [54]. To find this set of solutions, defining a way to compare different solutions in the multiobjective framework is relevant. The Pareto dominance establishes the relationship between two different solutions: for a solution x_1 to be better than solution x_2, it should be at least as good in all objectives and outperform in at least one of them [55].

Table 1
Comparison of feature selection approaches applied for microarray datasets.

Reference	Feature selection type				Neuroevolution	Multiobjective
	Filter	Wrapper	Embedded	Ensemble		
Maniruzzam et al. [13]	✓					
Gumeai et al. [14]	✓					
Jain et al. [23]		✓				
Pradana et al. [26]		✓				
Nurlaily et al. [28]		✓				
Garro et al. [5]		✓				
Garro et al. [29]		✓				
Santhakumar et al. [30]		✓				
Wang et al. [31]		✓				
Ahadzadeh et al. [32]		✓				
Turgut et al. [33]			✓			
Li et al. [34]			✓			
Abasabadi et al. [4]	✓	✓				
Pragadeesh et al. [17]	✓	✓				
Sarbazi–Azad et al. [16]	✓	✓				
Shukla et al. [18]	✓	✓				
Dashtban et al. [19]	✓	✓				
Pashaei et al. [21]	✓	✓				
Xiong et al. [22]	✓	✓				
Prasad et al. [25]	✓	✓				
Bir–Jmel et al. [27]	✓	✓				
Wang et al. [35]	✓			✓		
Hengpraprohm et al. [36]	✓			✓		
Trevizan et al. [37]	✓			✓		
Ghosh et al. [15]	✓	✓		✓		
Shukla [20]	✓	✓		✓		
Alrefai et al. [24]	✓	✓		✓		
Lai et al. [38]	✓	✓		✓		
Manikandan et al. [47]	✓	✓		✓		
Grisci et al. [7]			✓		✓	
Grisci et al. [8]	✓		✓		✓	
Rangasamy et al. [41]		✓				✓
Othman et al. [43]		✓				✓
Dussaut et al. [6]	✓	✓				✓
Sharma et al. [42]	✓	✓				✓
Dabba et al. [44]	✓	✓				✓
Xie et al. [48]	✓	✓				✓
Chaudhuri et al. [45]	✓	✓		✓		✓
Qing et al. [46]		✓	✓	✓		✓
Proposed	✓	✓			✓	✓

Evolutionary algorithms (EAs) are a popular approach for multiobjective optimisation. EAs are bio-inspired optimisation methods based on Darwin's evolution principles [56]. They are stochastic optimisation algorithms in which a population of solutions evolves through generations by crossover and mutation processes, searching to optimise their fitness. Like in natural selection, the fittest individuals would have a higher probability of passing their 'genes' onto the next generation.

3.2.1. $\mathscr{S}$-Metric selection evolutionary multiobjective algorithm (SMS-EMOA)

Among evolutionary multiobjective algorithms (EMOA) [57], there exists a category known as indicator-based methods [58]. These algorithms employ indicators to evaluate the quality of the Pareto front. Indicator-based methods aim to find Pareto-optimal solutions that optimise their indicator's value.

$\mathscr{S}$-metric selection evolutionary multiobjective algorithm (SMS-EMOA) is a popular indicator-based algorithm which uses the $\mathscr{S}$-metric [59], also known as hypervolume, as its indicator. The $\mathscr{S}$-metric computes the hypervolume between the Pareto front and a reference point; the latter is often set to be the worst possible outcome in each objective, meaning that a high hypervolume value is desired. The proximity, distribution and extension of the Pareto front influence the hypervolume.

SMS-EMOA follows an $\mu + 1$ framework, which means that the initial population is of size μ and in each iteration, one individual is generated and added to the population, and then, one individual is removed from the population to reduce it again to size μ. The reduction procedure is the distinctive step from SMS-EMOA. Alg. 1 describes this procedure. The first step is to sort the population into ranked fronts using the fast-non-dominated-sorting method introduced in the Non-Dominated Sorting Algorithm II (NSGA-II) [54] (Alg. 1, Line 1). The first front, $\mathscr{R}_1$, would consist of the set of non-dominated solutions from the population. For computing subsequent fronts, solutions from previous fronts are excluded. This sorting process continues until all the population members are assigned to a front. The second step from the Algorithm 1 aims to identify the solution from the worst ranked front, $\mathscr{R}_v$, that contributes the least to the hypervolume. Subsequently, this solution is removed from the population.

Algorithm 1 SMS-EMOA: reduce(P)

Input: Current population P
Output: Updated population
1: $\{\mathscr{R}_1, ..., \mathscr{R}_v\} \leftarrow$ fast-non-dominated-sorting(P)
2: $r \leftarrow \min_{s \in \mathscr{R}_v}[\Delta_{\mathscr{S}}(s, \mathscr{R}_v)]$
3: **return** $(P \backslash \{r\})$

3.3. Neuroevolution

Training ANNs can be described as an optimisation task, and as stated above, EAs are a suitable tool for solving them [60]. The research field of applying evolutionary computation methodologies to optimise, and in lesser degree to train, ANNs is commonly referred to as neuroevolution [60].

Some neuroevolution approaches include the optimisation of the network topology, which is an advantage over conventional gradient-based methods. Moreover, neuroevolution methods have shown competitive performance against contemporary algorithms while significantly reducing the learning time for tasks such as reinforcement learning benchmarks [61].

3.3.1. NEAT

One of the most popular algorithms in this field is Neuroevolution of Augmenting Topologies (NEAT) [49], which owns its popularity because it is one of the first methods to optimise both the ANN topology and its connection weight effectively.

NEAT execution starts with a population of random networks with minimal topologies, which means a fully connected network between the inputs and the outputs. Then, the population evolves through generations by adding complexity to their structures, allowing more complex behaviours. Furthermore, NEAT incorporates a speciation method that protects novel topologies to evolve and compete against the former individuals.

In EAs, genetic encoding is how an individual is represented to apply the crossover and mutation operators. In NEAT, each individual is an ANN, and it is represented as two arrays: one array containing all the neurons called node genes and one containing all the connections called connection genes. Each node gene has an identification value and is labelled according to whether they are inputs, outputs, or hidden neurons. The connection genes contain the information of its input and output nodes, the weight value, an enable flag, and an identification value referred to as the innovation number.

As aforementioned, the word 'gene' is used in EAs to describe one element of the genetic encoding. However, in future sections, the feature selection task of microarray datasets will be introduced in which each feature is related to a gene referring to a segment of the DNA. To avoid confusing the reader with the term 'gene', from now on, it will refer to its biological meaning, and the SMS-MONEAT node genes and connection genes will be called just nodes and connections.

Parents are selected to generate a new individual in NEAT, and then the evolutionary operators are performed. There is a specific probability of executing each of them. The evolutionary operators are explained as follows:

- **Crossover**: The crossover operator uses the 'innovation number' to find matching connections between parents. If both parents have the same fitness, the new individual inherits all the connections. Otherwise, the matching connections between both parents and the unmatching connections from the fitter parent are inherited.
- **Mutation - evolve weight**: A weight can be mutated by disturbing its value or changing it for a newly generated random value.
- **Mutation - add a new node**: a new node can be added by disabling an existing connection and creating two new connections to this node.
- **Mutation - add a new connection**: a new connection can be added between two existing nodes with a random weight value.

3.3.2. FS-NEAT and N3O

Feature Selective NEAT (FS-NEAT) is a variant of NEAT [40], which includes a feature selection process while optimising the ANNs, both weights and topology. The main difference between NEAT and FS-NEAT is how the population is initialised. While NEAT initialises each individual as a fully connected network between the input and output nodes, FS-NEAT initialises them with just one random connection between an input and output. Then, it relies on the evolutionary operators to create new paths between inputs and outputs to approach the feature selection task.

Algorithm 2 SMS-MONEAT

Input: SMS-MONEAT hyperparameters and a dataset to optimise the ANNs
Output: A population of ANNs P

```
P_0 ← initialise()
t ← 0
while t ≠ t_max do
    y ← generate_offspring(P_t)
    P_{t+1} ← reduce(P_t ∪ {y})
    t ← t + 1
end while
```

Furthermore, N3O [8] is a FS-NEAT variant. Its name stands for '3 new operators', and as its name suggests, three main changes were made: two new structural mutation operators and a modification to the crossover operator. They are described as follows:

- **Mutation - add a guided input**: a new input is selected using a probability value and added to the network. Each input probability was defined using the p-value of the Kruskal Wallis H-test.
- **Mutation - swap an input**: a new input can be added by swapping it for an existing input.
- **Modified crossover**: the crossover includes a new probability of adding unmatched connections from the less fit parent to the offspring.

N3O was the main inspiration behind the design of the multiobjective algorithm proposed in this work.

4. Proposed approach

This section describes the neuroevolution-based multiobjective algorithm named SMS-MONEAT proposed in this paper. It also elaborates on including a new speciation method within an external archive to enhance the diversity of the selected feature subset within SMS-MONEAT. Additionally, it outlines the proposed gene selection and microarray classification pipeline, incorporating SMS-MONEAT as a wrapper method.

4.1. $\mathscr{S}$-Metric selection - multiobjective neuroevolution of augmenting topologies (SMS-MONEAT)

The $\mathscr{S}$-metric selection - multiobjective neuroevolution of augmenting topologies (SMS-MONEAT) combines SMS-EMOA framework with N3O evolutionary operators. The main procedure is outlined in Alg. 2. SMS-MONEAT starts by generating a random population with just one input and output linked to each other, as in N3O. Then, in each iteration, a new individual is generated using N3O operators and added to the population. After this, the population is reduced by removing the individual with the least hypervolume contribution, as in SMS-EMOA. By using N3O mechanisms, SMS-MONEAT can perform a feature selection process during the ANNs optimisation. The algorithm's hyperparameters will be described in Section 5.2.

Fig. 1 illustrates the process of generating offspring within SMS-MONEAT. First, two individuals are selected as parents from the population. Then, the crossover and mutation operators are applied to generate the offspring. In this example, the crossover and all mutation operators are applied to demonstrate their effects. However, each operator has a defined probability of being applied, which is set as a hyperparameter. Additionally, there is no specific order for applying the mutation operators within the algorithm, but the flowchart follows the way the algorithm was implemented. For further information on each of the evolutionary operators, see Section 3.3.1.

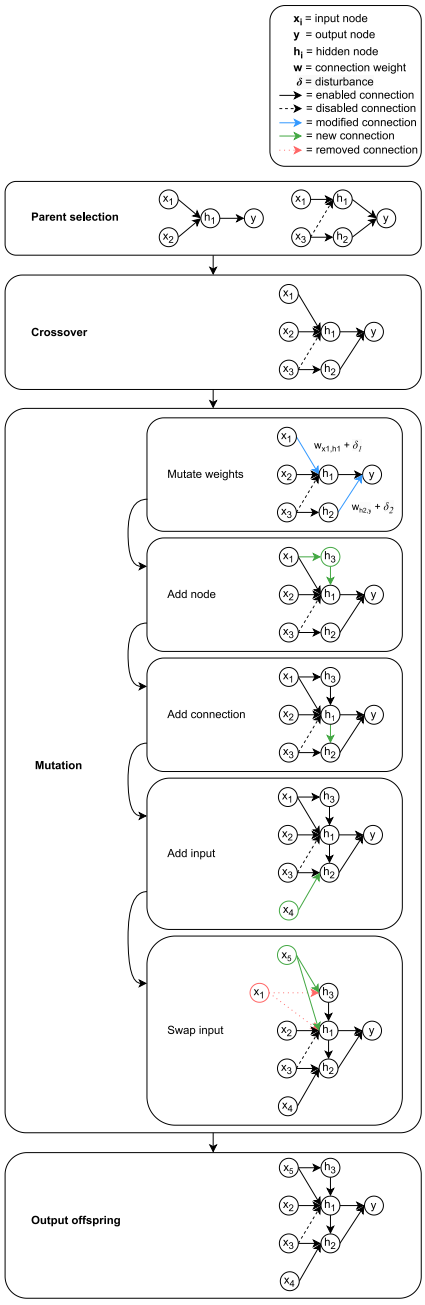


Fig. 1. Flowchart with an example of the offspring generation within SMS-MONEAT.

4.2. Selected features based speciation archive

As stated before, microarray datasets have high dimensional data, and training a classification model that generalises well on new data may be challenging. Therefore, a speciation method was proposed to divide species according to the subsets of features selected and stored in an external archive [62,63]. Alg. 3 shows the SMS-MONEAT main procedure, including an external archive for the speciation method.

When adding a new individual to the external archive, it must be compared against the existing species. If this new individual does not belong to an existing species, a new species is created. All species have a maximum number of individuals to avoid that one species suppressing diversity in the external archive. To remove one individual from an overpopulated species, only the loss function is used as an indicator to decide the less-fit individual since all species members have the same number of features selected. Furthermore, the external archive also has a maximum number of individuals. When this value is exceeded, an individual must be removed. To do so, the less fit individuals of each species are grouped, and the one with the lowest hypervolume contribution is removed. Alg. 4 describes the procedure to add a new individual to the external archive, while Alg. 5 shows how an individual is removed.

Algorithm 3 SMS-MONEAT with speciation method

Input: SMS-MONEAT hyperparameters and a dataset to optimise the ANNs
Output: A population of ANNs P and an archive population Q

$P_0 \leftarrow$ initialise()
$Q_0 \leftarrow \emptyset$
for each $p \in P_0$ **do**
  $Q_0 \leftarrow$ add_to_archive(Q_0, p)
end for
$t \leftarrow 0$
while $t \neq t_{max}$ **do**
  $y \leftarrow$ generate_offspring()
  $P_{t+1} \leftarrow$ reduce($P_t \cup \{y\}$)
  $Q_{t+1} \leftarrow$ add_to_archive(Q_t, y)
  $t \leftarrow t + 1$
end while

Algorithm 4 add_to_archive(Q, x)

Input: Current archive population Q split by species, a new individual x, a maximum archive population size ψ and a maximum size of each species σ.
Output: Updated archive population

$added \leftarrow False$
for each $S \in Q$ **do** ▷ For each species in Q.
  if $x_{features} = S_{features}$ **then**
    $S \leftarrow S \cup \{x\}$
    if size(S) $> \sigma$ **then**
      $z \leftarrow \text{argmin}_{s \in S}(s_{fitness})$
      $S \leftarrow (S \backslash \{z\})$
    end if
    $added \leftarrow True$
  end if
end for
if $added = False$ **then**
  $S \leftarrow \{x\}$ ▷ Create a new species.
  $Q \leftarrow Q \cup \{S\}$
end if
if size(Q)$> \psi$ **then**
  $Q \leftarrow$ reduce_archive(Q)
end if
return Q

Algorithm 5 reduce_archive(Q)

Input: Current archive population Q split by species
Output: Updated archive population

$A \leftarrow \emptyset$
for each $S \in Q$ **do**
  $z \leftarrow \text{argmin}_{s \in S}(s_{fitness})$
  $A \leftarrow A \cup \{z\}$
end for
$\{\mathscr{R}_1, ..., \mathscr{R}_v\} \leftarrow$ non-dominated-sorting(A)
$r \leftarrow \text{argmin}_{s \in \mathscr{R}_v}[\Delta_{\mathscr{S}}(s, \mathscr{R}_v)]$
return $(Q \backslash \{r\})$

4.3. Feature selection and microarray classification

A methodology for feature selection and binary classification on microarray datasets was implemented, which included SMS-MONEAT as a wrapper method for feature selection. The methodology consists of the following steps:

1. A feature selection process based on a filter method using the Kruskal Wallis H-test.
2. Scaling the data using the MinMaxScaler function.
3. A feature selection process and a binary classification model based on a wrapper method using the SMS-MONEAT algorithm.
4. An external archive based on the feature-based speciation method.
5. A decision-making procedure to choose one solution from the final populations (main and archive populations).

Each of these steps will be explained in further detail.

4.3.1. Feature selection by statistical filter

A filter method was used as the first step of feature reduction. This allows to reduce the search space by removing most of the irrelevant features before executing the wrapper method. The Kruskal Wallis H-test was used to evaluate each feature, and the p-value was used as the threshold to determine whether a feature was relevant. During our experiments, features with a p-value lower than 0.01 were considered relevant, and the rest were discarded. The p-value for each feature is stored since it will also be applicable during the SMS-MONEAT execution since it is used by the mutation operator, which adds new inputs guided by this value. This step follows the example of the original methodology of N3O [8].

4.3.2. Scaling the data

To scale the data to values between zero and one, the MinMaxScaler function was used, described in Eqs. (3) [64]. First, Eq. (3a) computes the minimum and maximum values for each feature j in the dataset X and uses them to normalise each sample (i) value X_{ij}. Then, in Eq. (3b), the scaled value X'_{ij} are scaled between a lower ($\alpha \in \mathbb{R}$) and an upper ($\beta \in \{x \mid x \in \mathbb{R}, \alpha < x\}$) value limit, which were set to 0 and 1, respectively, in our experiments.

$$X'_{ij} = \frac{X_{ij} - \min(X_j)}{\max(X_j) - \min(X_j)}, \tag{3a}$$

$$z_{ij} = \alpha + X'_{ij}(\beta - \alpha). \tag{3b}$$

4.3.3. Feature selection by wrapper method based on SMS-MONEAT

The primary purpose of using SMS-MONEAT as a wrapper method is to minimise the number of features selected while maximising the performance of the ANNs in the classification task. Therefore, the fitness functions consist of two objectives: a binary cross entropy loss function and the number of features selected scaled by a coefficient value. Trying to scale both objectives is relevant since SMS-MONEAT relies on the hypervolume indicator. The Eq. (4) describes the binary cross-entropy function, presented alongside N3O [8].

$$g(a, y) = \frac{1}{Q}\sum_{q \in Q} -\frac{1}{n^q}\sum_{i=1}^{n^q}\left[y_i \ln a_i + (1 - y_i)\ln(1 - a_i)\right] \tag{4a}$$

$$+ \frac{\lambda}{2n}\frac{1}{c}\sum_{k=1}^{c}\omega_k^2, \tag{4b}$$

which is divided in two main terms: Eq. (4a) is the binary cross-entropy term while Eq. (4b) a regularisation term. Q is the set of classes, q represents each of these classes, $n^q \in \mathbb{N}^+$ the number of samples of q, y_i and a_i are the true label value and the predicted value for the sample i respectively, $\lambda \in \mathbb{R}$ is a regularisation parameter, $n \in \mathbb{N}^+$ the number of samples, $c \in \mathbb{N}$ the number of enabled connections and $\omega_k \in \mathbb{R}$ the weight value of the connection k.

The fitness function is described in the Eq. (5). The value of $\alpha \in \mathbb{R}$ is the coefficient to scale the number of features selected. In our experiments, α was set equal to 0.1.

$$fitness = [g(a, y), \alpha f_s]. \tag{5}$$

The algorithm's output is a population of ANNs that might have different topologies and different subsets of features selected. Furthermore, the feature-based speciation method described in Section 4.3 generates an additional output population and is stored in an external archive.

During SMS-MONEAT execution, connections are enabled or disabled, which might cause invalid ANNs. Therefore, it is necessary to define a method to evaluate invalid individuals so that they do not affect the exploration of the algorithm. In our method, invalid individuals were assigned ∞, so they were discarded in the reduce mechanism (Alg. 2 line 5).

4.3.4. Decision-making procedure

To choose one ANN model from the final population output by SMS-MONEAT, the models were evaluated in the training dataset as well as in a validation dataset to observe how well they generalise to new data. Two indicators were used: the binary cross entropy (Eq. (4)) and the geometric mean (Eq. (6)).

$$gm = \sqrt{TP_{rate} * TN_{rate}}. \tag{6}$$

The geometric mean (gm) is computed as shown in Eq. (6) using the rate of true positives ($TP_{rate} \in \{x \mid x \in \mathbb{R}, 0 \le x \le 1\}$) and true negatives ($TN_{rate} \in \{x \mid x \in \mathbb{R}, 0 \le x \le 1\}$).

This indicator was used to avoid bias within the imbalanced datasets. After evaluating each model with both indicators and in both datasets, a weighted sum was computed to select one model. The Eq. (7) describes the weighted sum used,

$$z = \omega_1[g(a_{train}, y_{train}) + g(a_{val}, y_{val})] + \omega_2(gm_{train} + gm_{val}), \tag{7}$$

where z is the weighted sum outcome for a certain individual in the population and is computed using the results from the loss function $g(a, y)$ and the geometric mean gm. The suffix $train$ is related to the evaluation of the training dataset, while the suffix val refers to the evaluation of the validation dataset. The weights used for each of the terms are represented in $\omega_1 \in \{x \mid x \in \mathbb{R}, 0 \le x \le 1\}$ and $\omega_2 \in \{x \mid x \in \mathbb{R}, 0 \le x \le 1\}$. Their values used within this work are provided in Table 5.

5. Experimental setup

In this section, we detailed the experiments carried out within this work to assess SMS-MONEAT performance on the feature selection and microarray classification tasks.

A stratified 10-fold cross-validation with three repetitions was used for SMS-MONEAT experiments. Hence, the algorithm was executed 30 times in each of the 20 benchmark datasets within the cross-validation process. During the cross-validation process, seven folds were employed for the optimisation process (training); two folds were used as a validation subset for selecting one solution from the population (please refer to Section 4.3.4), and the last fold was used for evaluation. The flowchart diagram of the experiment pipeline is shown in Fig. 2 to complement our explanation.

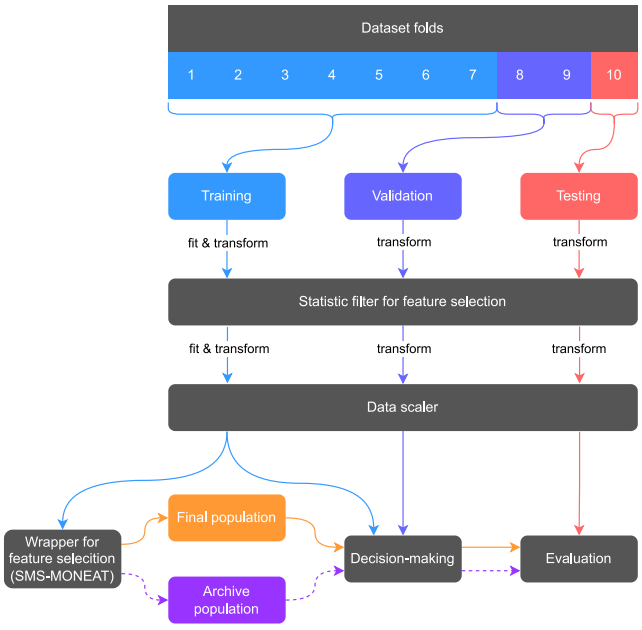


Fig. 2. SMS-MONEAT experiment diagram for feature selection and microarray classification.

5.1. Comparison with other algorithms

SMS-MONEAT performance was compared against several state-of-the-art algorithms. For instance, it was compared against N3O, described in Section 3.3.2. The comparison against N3O also works as an ablation study since it is possible to observe the advantages of including a multiobjective framework in SMS-MONEAT. The pipeline shown in Fig. 2 was also used with N3O as the wrapper method, cutting out the external archive population. Nevertheless, since N3O is a single objective optimisation algorithm, it only employs the binary cross-entropy function (Eq. (4)) as an optimisation function.

In addition to that, a multiobjective approach using SMS-EMOA was also employed to compare against SMS-MONEAT. This method used a binary codification, meaning that a binary array sets which features are selected or discarded, and employed $k-$Nearest Neighbours (KNN) as a classification model. This also worked as an ablation study since it is possible to compare the difference between the binary codification and a neuroevolution codification such as the one used in SMS-MONEAT.

Three more state-of-the-art algorithms were selected to compare against SMS-MONEAT: MOCHC [65], the simple, fast and efficient feature selection algorithm (SFE) [32] and SFE-PSO. MOCHC showed an outstanding performance in the study carried out by Dussaut et al. [6], where the performance of multiple multiobjective algorithms was compared on feature selection and microarray classification tasks. On the other hand, SFE is a feature selection algorithm designed for high-dimensional data, and SFE-PSO is a hybrid version of SFE and PSO. Both SFE and SFE-PSO have been with microarray datasets [32]. For all of these algorithms, KNN was used as a classification model.

A particular difference between neuroevolution-based methodologies (SMS-MONEAT and N3O) and the rest of the algorithms (SMS-EMOA, MOCHC, SFE and SFE-PSO) is that in the former ones, the classification model (an ANN) is optimised during the feature selection process and, during the fitness evaluation only the prediction is executed. In contrast, in the latter ones, the classification model (KNN) needs to be trained in each fitness evaluation. Furthermore, to ensure robustness when evaluating the KNN model, a 3-fold cross-validation was executed. Hence, there is a difference in the computational cost between evaluating a subset of features in the neuroevolution-based methodologies and the rest of the algorithms. To simplify the analysis, we assumed that computing one fold of the cross-validation for KNN was equivalent to evaluating an ANN model generated by the neuroevolution-based methods.

For SMS-EMOA, MOCHC, SFE and SFE-PSO, the pipeline shown in Fig. 2 was modified. Since they perform cross-validation when evaluating the KNN model, there was no need for a validation subset outside the wrapper; therefore, nine folds were used as training datasets for the wrapper method and the last fold to evaluate its output. Furthermore, similarly to SMS-MONEAT, each of the other algorithms was executed 30 times within the cross-validation process on each dataset. This was done to ensure reproducibility and perform a statistical analysis of the outcome.

5.2. Hyperparameters

EAs performance relies on multiple hyperparameters, and SMS-MONEAT is no exception. iRace [10], an automatic configuration methodology, was employed to find the best set of hyperparameters for SMS-MONEAT on the feature selection and microarray classification tasks. iRace is an extension of the racing methodology F-Race [66], which uses the Friedman test to compare different sets of hyperparameters and, using a *post hoc* test, discard those configurations with a lower performance.

iRace was used to tune the parameters related to the evolutionary operators. It performed 1000 executions of SMS-MONEAT, testing different hyperparameter configurations. Five microarray datasets from the Curated Microarray Database (CuMiDa) were employed during the tuning process [67]. Each dataset is related to a different type of cancer: colon, leukemia, liver, Breast and prostate. Further information on these datasets is shown in Table 2. The hypervolume was used as the target to optimise the hyperparameters, computed using the two objectives from Eq. (5).

Additionally, iRace was used to tune the hyperparameters of the other algorithms to which SMS-MONEAT was compared. SMS-EMOA and MOCHC followed the same setup as SMS-MONEAT by using the hypervolume of the population as the target function. For N3O, SFE and SFE-PSO, the binary cross entropy (Eq. (4)) was used as the target optimisation function since these algorithms are single-objective optimisation.

Table 3 shows the tuned six hyperparameters (the crossover and all of the mutation operators probabilities) from SMS-MONEAT and N3O and the settings used on iRace for each. The start value for each parameter was selected based on N3O [8]. The minimum and maximum values were set equally for all mutation operators to observe their influence on the algorithm performance. Moreover, Table 3 shows the best four configurations found by iRace for SMS-MONEAT and N3O. There are some observable patterns from the top configurations for each algorithm.

Regarding SMS-MONEAT top configurations obtained by iRace, the crossover probabilities seemed to approach the minimum value set. This could be related to invalid individuals that can be generated within this operator. On the other hand, the structural mutation operators tend to approach the maximum value, allowing for more exploration. Finally, for the weight mutation, the probabilities were closer to the initial value, which is lower than the structural mutations' probabilities and could mean that the mutation of weights has a lower positive influence on the algorithm performance.

For the case of N3O, the crossover probabilities approached the minimum value set in 3 out of 4 configurations, similar to SMS-MONEAT configurations. However, N3O's third configuration got a much higher value than the rest. This configuration also has a higher probability of adding a new input and a lower probability of adding a new connection than the rest of the N3O configurations. We argue that for this particular configuration, the crossover operator got less penalised since a lower number of new connections were created. Therefore, the number

Table 2
Datasets used as instances for SMS-MONEAT automatic configuration using iRace.

Dataset	Cancer Type	Samples	Features	Class distribution [%]
GSE42568	Breast	116	54 675	87/13
GSE8671	Colon	63	54 675	51/49
GSE46602	Prostate	49	54 675	71/29
GSE57957	Liver	75	47 323	52/48
GSE71935	Leukemia	46	54 675	80/20

Table 3
iRace setup for automatic configuration of SMS-MONEAT and N3O. Experiment setup (Columns 2–4) and top four configurations for SMS-MONEAT (Columns 5–8) and N3O (Columns 9–12).

Hyperparameter	Settings			SMS-MONEAT top configurations				N3O top configurations			
	Start	Min	Max	1st	2nd	3rd	4th	1st	2nd	3rd	4th
Crossover	0.75	0.50	1.00	**0.5662**	0.5198	0.5803	0.5234	**0.5925**	0.5774	0.8604	0.5678
New input	0.05	0.00	0.15	**0.1226**	0.1283	0.1443	0.1035	**0.0016**	0.0175	0.1391	0.0106
Swap input	0.05	0.00	0.15	**0.0972**	0.1126	0.1205	0.1189	**0.1239**	0.1358	0.1326	0.1483
New connection	0.05	0.00	0.15	**0.1352**	0.1353	0.1425	0.1164	**0.0874**	0.0922	0.0569	0.1154
New node	0.03	0.00	0.15	**0.1390**	0.1489	0.1369	0.1215	**0.1339**	0.1394	0.1332	0.1421
Mutate weight	0.04	0.00	0.15	**0.0498**	0.0501	0.0450	0.0613	**0.0980**	0.1081	0.1251	0.1185

of invalid individuals generated from the crossover was reduced. N3O configurations tend to have a higher probability of swapping inputs than adding new ones. This is a surprise since the number of features selected was not considered within the optimisation function as in SMS-MONEAT.

Table 4 shows the iRace setup and top configurations obtained for SMS-EMOA, MOCHC, SFE and SFE-PSO. Generally, consistency was found between the top 3 configurations for each algorithm.

For SMS-EMOA, the best configuration was the starting configuration, with values in the middle between the minimum and maximum values set for the experiment. The second and third top configurations have a lower crossover probability but a higher mutation probability. For this algorithm, the mutation operator plays a crucial role in exploring new features. On the other hand, the mutation operator also tends to increase the number of features selected. This may explain why the best configuration was found at the midpoint, where a balance between exploration and minimising the number of features was achieved.

For MOCHC, the initial convergence count approximates 0.01, which was the minimum value that was set. This might be related to the fact that the convergence count is used as the threshold to allow two individuals to perform a crossover, and since the algorithm aims to minimise the number of features selected, having a smaller threshold would allow promising individuals to participate in the evolutionary process. The best individual has a larger mutation probability than the 2nd and 3rd configurations; this might help the algorithm to escape from local optimum more effectively.

For SFE, the three top configurations had a minimum value for the non-selection operator of 0.1, which is 10% of the features. For the non-selection operator maximum value, the first and third top configurations got a similar value close to half of the features. In contrast, for SFE-PSO the non-selection operator maximum value is lower, on average 35%. This makes sense since SFE have a more significant dependency on this parameter than SFE-PSO.

The remaining hyperparameters for each algorithm are presented in Table 5. These hyperparameter values were used within the iRace tuning and in further experiments on microarray datasets.

5.3. Datasets

Using benchmark problems allows the research community to assess, validate and explain many evolutionary algorithms. In this study, we have opted for established, curated, and validated benchmark datasets from prior research [67–70] to (i) evaluate the algorithms employed in this work, (ii) employ clearly defined indicators for method-to-method comparisons, (iii) facilitate the explanation of why a particular method outperforms others, and (iv) derive robust conclusions from the results presented in the subsequent sections. Thus, the proposed algorithm was analysed using multiple microarray datasets for binary classification, most of them gathered from CuMiDa [67], with additional datasets obtained from other studies [68–70]. Table 6 provides details on the 20 datasets employed in this research, all associated with five different cancer types: colon, leukemia, liver, breast, and prostate. Datasets prefixed with 'GSE' are part of the CuMiDa database. To simplify dataset nomenclature, each was assigned a new code name (Column 2) indicating the corresponding cancer type. These datasets exhibit high dimensionality, averaging 114.85 samples and 30 332.45 features. Additionally, some datasets feature imbalanced class distributions, with Breast_1 displaying the most pronounced imbalance (93% for one class), as indicated in Column 5.

5.4. Development framework

The experiments were developed using Python 3, and several libraries were employed. For example, Scikit-Learn [64] was used for processing the datasets and the main structure of the experiments, PyTorch [71] was used for computing the ANNs generated by SMS-MONEAT, and PyGMO [72] was used for computing the hypervolume. The code can be found in the following GitHub link: https://github.com/dan-gn/SMS-MONEAT.git. The experiments were performed on an Intel i7-6500U processor with 16 GB of RAM.

6. Discussion of results

The proposed methodology used for feature selection and microarray classification consisted of two sequential steps: a filter method for feature selection based on the Kruskal Wallis H-test (KW) and a wrapper method in which different algorithms were tested and compared. The results of each step are presented separately as follows.

6.1. Feature selection using the Kruskal–Wallis H test as filter method

Using a filter method, as discussed in Section 4, for feature selection before executing a wrapper method could benefit its performance by removing most irrelevant features and, therefore, reducing its search space. Table 7 shows the mean and standard deviation (±) of selected features obtained by the filter method in each dataset and in average from all the datasets. On average, 75.31% of the features were removed during this step (see average of original features and selected features in Table 7).

Table 4
iRace setup for automatic configuration of SMS-EMOA, MOCHC, SFE and SFE-PSO. Experiment setup (Col. 2–4) and top three configurations (Col. 5–8).

Hyperparameter	Settings			Top configurations		
	Start	Min	Max	1st	2nd	3rd
SMS-EMOA						
Crossover	0.75	0.50	1.00	**0.7500**	0.5767	0.6132
Mutation	0.10	0.00	0.20	**0.1000**	0.1354	0.1433
MOCHC						
Initial Convergence Count	0.02	0.01	0.10	**0.0110**	0.0110	0.0121
Mutation	0.35	0.05	0.50	**0.3976**	0.1055	0.1046
SFE						
Non-selection rate – Min	10^{-3}	10^{-1}	10^{-5}	$\mathbf{10^{-1}}$	10^{-1}	10^{-1}
Non-selection rate – Max	0.30	0.10	0.50	**0.44559**	0.2318	0.4692
Selection mode – N. of features	1	1	10	**2**	5	2
SFE-PSO						
Non-selection rate – Min	10^{-3}	10^{-1}	10^{-5}	$\mathbf{10^{-1}}$	10^{-2}	10^{-1}
Non-selection rate – Max	0.30	0.10	0.50	**0.3884**	0.2738	0.3880
Selection mode – N. of features	1	1	10	**7**	10	8
Inertia weight	0.01	10.00	10.00	**7.6267**	3.3603	6.9857
Cognitive coefficient	0.01	10.00	10.00	**8.4753**	8.4953	7.7521
Social coefficient	0.01	10.00	10.00	**5.8493**	2.0704	7.7384

Table 5
Hyperparameter values used during microarray experiments.

Hyperparameter	Value
Shared	
Number of evaluations	18 000
SMS-MONEAT & N3O	
Population size	100
Parent selection	Binary tournament
Crossover and mutation probabilities	Optimised by iRace[a]
Adding a connection from the less fit parent	0.50
Adding an unmatching connection when same fitness	0.80
Disabling a connection when disabled in one parent	0.75
Polynomial mutation disturbance var. (weight mut.)	5
Changing to random value (weight mut.)	0.10
Lower and upper limits for weights on initialisation	$[-1, 1]$
Regularisation parameter from the loss function (λ)	0.5
Aggregation function	mean
Activation function (input & hidden nodes)	$tanh(2.45x)$
Activation function (output node)	$exp\frac{-5x}{2}$
Decision-making [ω_1, ω_2]	[0.35, −0.15]
SMS-EMOA	
Population size	100
Parent selection	Binary tournament
Crossover	One-point
Mutation	Binary
Crossover and mutation probabilities	Optimised by iRace[a]
MOCHC	
Population size	100
Convergence value	1
Preserved population on reset	0.05
Mutation probability	Optimised by iRace[a]
Initial Convergence Count	Optimised by iRace[a]
SFE & SFE-PSO	
Population size	1
Non-selection operator rate range	Optimised by iRace
Number of features selected on selection mode	Optimised by iRace[a]
SFE-PSO	
Population size	100
PSO transition evaluation threshold	2000
Inertia weight, cognitive and social coefficients	Optimised by iRace[a]

[a] Hyperparameters optimised by iRace are shown in the Tables 3 and 4. The best configuration for each algorithm was used.

Table 6
Microarray datasets used within SMS-MONEAT performance testing. Dataset with *GSE* prefix were gathered from CuMiDa [67]. The second column sets a code name for each dataset, which relates them to their cancer type and simplifies the analysis of the results in future sections.

Dataset	Code name	Samples	Features	Class distribution [%]
GSE22820	$Breast_1$	139	33 579	93/07
GSE59246	$Breast_2$	101	36 622	55/45
GSE70947	$Breast_3$	289	35 981	51/49
Van De Vijver et al. [69]	$Breast_4$	97	24 481	53/43
GSE25070	$Colon_1$	52	24 526	50/50
GSE32323	$Colon_2$	33	54 675	52/48
GSE44076	$Colon_3$	194	49 386	50/50
GSE44861	$Colon_4$	105	22 277	50/50
GSE22529_U133A	$Leukemia_1$	52	22 283	79/21
GSE22529_U133B	$Leukemia_2$	52	22 645	79/21
GSE33615	$Leukemia_3$	71	33 579	70/30
GSE63270	$Leukemia_4$	101	54 675	59/41
Golub et al. [68]	$Leukemia_5$	72	7129	65/35
GSE14520_U133A	$Liver_1$	357	22 277	51/49
GSE50579	$Liver_2$	76	36 547	84/16
GSE62232	$Liver_3$	91	54 675	89/11
GSE6919_U95Av2	$Prostate_1$	124	12 625	50/50
GSE6919_U95B	$Prostate_2$	124	12 620	52/48
GSE11682	$Prostate_3$	31	33 467	52/48
Singh et al. [70]	$Prostate_4$	136	12 600	57/43

Table 7
Mean and standard deviation (±) of the number of features selected by the filter method based on the Kruskal Wallis H-test (KW).

Dataset	Original features	Selected features (KW)
$Breast_1$	33 579	7 526.73 ±308.25
$Breast_2$	36 622	4 185.67 ±807.20
$Breast_3$	35 981	17 322.87 ±517.40
$Breast_4$	24 481	1 051.47 ±419.88
$Colon_1$	24 526	5 668.67 ±551.84
$Colon_2$	54 675	13 464.80 ±1717.52
$Colon_3$	49 386	27 907.30 ±254.99
$Colon_4$	22 277	5 504.00 ±1121.00
$Leukemia_1$	22 283	2 214.63 ±478.68
$Leukemia_2$	22 645	1 794.30 ±612.03
$Leukemia_3$	33 579	10 031.83 ±267.23
$Leukemia_4$	54 675	14 493.73 ±832.36
$Leukemia_5$	7129	892.37 ±143.13
$Liver_1$	22 277	14 516.00 ±334.47
$Liver_2$	36 547	3 450.50 ±1084.14
$Liver_3$	54 675	15 200.67 ±1190.94
$Prostate_1$	33 467	1 479.07 ±435.39
$Prostate_2$	12 625	1 019.13 ±185.53
$Prostate_3$	12 620	616.33 ±147.29
$Prostate_4$	12 600	1 845.47 ±730.49
Average	30 332.45	7 488.48 ±598.75

6.2. Wrapper algorithms performance comparison

SMS-MONEAT was employed on the feature selection and microarray classification task, and its performance was compared against N3O, SMS-EMOA, MOCHC, SFE and SFE-PSO. Different indicators were used to evaluate their performance, such as:

1. The binary cross-entropy (Eq. (4)) from the individual chosen by the decision-making procedure evaluated in the testing dataset.
2. The number of selected features from the individual chosen by the decision-making procedure.
3. The geometric mean (Eq. (6)) from the individual chosen by the decision-making procedure evaluated in the testing dataset.
4. The hypervolume of the population using the two terms from the fitness function (Eq. (5)).
5. The execution time for each algorithm.

Note that the first two aforementioned indicators, the binary cross entropy (Point 1) and the number of features selected (Point 2), are the two objectives that compose the fitness function (Eq. (5)), which SMS-MONEAT aims to optimise. These two indicators were also used to compute the hypervolume (Point 4).

SMS-EMOA and MOCHC used the same objective function as SMS-MONEAT. On the other hand, N3O, SFE and SFE-PSO just used binary cross entropy (Point 1) as an optimisation function, given that they are a mono-objective optimisation algorithm.

6.2.1. Number of selected features

Table 8 shows the number of selected features by each algorithm. From SMS-MONEAT, two different solutions were obtained: one from the final population referred to as $SMS\text{-}MONEAT_P$, and one from the external archive, named $SMS\text{-}MONEAT_Q$. The last row shows the average from all of the experiments performed. The cell with the best average per dataset (row) was highlighted in orange. $SMS\text{-}MONEAT_P$ obtained the best performance on average by selecting 2.88 features from all experiments. In all datasets, $SMS\text{-}MONEAT_P$ got a slightly lower number of features than $SMS\text{-}MONEAT_Q$ (except on $Liver_2$ where both got the same mean value), which was expected since $SMS\text{-}MONEAT_Q$ employs the speciation method which aims to maintain the diversity of the subset of selected features, prioritising the diversity over reducing the number of features.

On the other hand, SFE had the highest average number of selected features compared with the rest of the algorithms, followed by SFE-PSO. This was expected since both are mono-objective optimisation algorithms and do not include the number of features as a function to minimise. N3O is also a mono-objective algorithm with significantly fewer selected features on average. This might be related to the fact that N3O starts with only one selected feature and progressively adds new features during optimisation. In contrast, SFE algorithms start with all features selected and remove features during their execution. SFE algorithms might converge to local optimums before reducing the number of features to values closer to the rest of the algorithms.

In addition, the Friedman's post hoc test was computed to compare the algorithms with a threshold p-value of 0.05. Fig. 3 shows the results from the statistical test. Each column of the figure represents the results for each algorithm against (A) $SMS\text{-}MONEAT_P$, (B) $SMS\text{-}MONEAT_Q$, (C) N3O, (D) SMS-EMOA, (E) MOCHC, (F) SFE and (G) SFE-PSO. Green and red squares represent a significant difference in performance, positive and negative, respectively. Yellow squares represent that no significant difference was found between both algorithms. Using the Friedman's

Table 8
Mean and standard deviation (±) of each experiment's number of selected features (lower values are better ↓). The last row describes the average value from all of the experiments. The best mean value obtained by an algorithm in each dataset is highlighted.

Dataset	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O [8]	SMS-EMOA [59]	MOCHC [65]	SFE [32]	SFE-PSO [32]
Breast$_1$	2.90 ± 1.3	3.13 ± 1.3	13.37 ± 14.9	2.07 ± 0.3	2.03 ± 0.2	411.00 ± 194.2	309.17 ± 165.4
Breast$_2$	3.67 ± 1.5	4.17 ± 1.4	11.13 ± 4.5	6.03 ± 3.3	34.63 ± 6.2	513.33 ± 122.9	379.53 ± 77.8
Breast$_3$	2.47 ± 1.2	2.90 ± 1.2	7.20 ± 3.8	5.13 ± 3.0	116.47 ± 10.0	332.77 ± 569.5	704.87 ± 477.0
Breast$_4$	3.23 ± 1.7	3.47 ± 1.5	9.23 ± 4.2	6.20 ± 3.0	14.70 ± 3.6	178.20 ± 24.4	130.53 ± 19.9
Colon$_1$	2.53 ± 1.2	3.07 ± 1.6	13.80 ± 14.3	3.07 ± 1.4	5.70 ± 2.0	389.27 ± 184.3	322.73 ± 123.3
Colon$_2$	2.00 ± 1.2	2.17 ± 0.9	16.00 ± 16.2	1.33 ± 0.5	3.73 ± 0.7	521.57 ± 367.9	385.93 ± 283.5
Colon$_3$	2.10 ± 1.0	2.37 ± 1.0	11.50 ± 20.7	2.63 ± 1.0	37.67 ± 19.8	1796.90 ± 491.6	1188.73 ± 451.5
Colon$_4$	2.63 ± 0.9	2.90 ± 1.1	9.67 ± 5.2	4.47 ± 2.0	36.47 ± 4.7	412.90 ± 168.1	216.20 ± 181.8
Leukemia$_1$	4.27 ± 10.4	4.37 ± 10.2	8.97 ± 7.6	2.23 ± 1.5	2.03 ± 0.5	136.33 ± 77.6	133.93 ± 61.2
Leukemia$_2$	3.07 ± 1.3	3.50 ± 3.5	9.07 ± 5.3	1.67 ± 0.8	1.83 ± 0.7	145.70 ± 90.3	102.53 ± 61.7
Leukemia$_3$	2.53 ± 1.1	2.87 ± 1.3	11.90 ± 17.6	1.57 ± 0.5	1.87 ± 0.6	475.60 ± 218.5	304.07 ± 112.3
Leukemia$_4$	3.47 ± 6.1	4.23 ± 6.7	7.97 ± 4.8	3.07 ± 1.5	10.73 ± 2.4	1179.27 ± 285.7	814.40 ± 159.3
Leukemia$_5$	3.30 ± 1.3	3.53 ± 1.1	10.03 ± 6.6	3.70 ± 2.0	3.97 ± 0.8	152.60 ± 17.3	106.97 ± 26.8
Liver$_1$	2.10 ± 0.9	2.33 ± 0.9	6.73 ± 4.7	8.80 ± 3.1	91.27 ± 8.6	1242.13 ± 17.9	891.03 ± 46.4
Liver$_2$	2.83 ± 1.2	2.83 ± 1.2	8.40 ± 3.8	4.40 ± 2.1	6.17 ± 1.7	394.40 ± 123.7	295.00 ± 93.3
Liver$_3$	1.53 ± 1.0	2.13 ± 1.7	9.50 ± 6.4	1.87 ± 0.5	6.23 ± 1.9	762.47 ± 411.4	640.77 ± 159.4
Prostate$_1$	3.53 ± 1.6	3.83 ± 1.7	10.40 ± 3.6	6.23 ± 3.3	18.47 ± 3.4	192.50 ± 44.1	135.20 ± 33.9
Prostate$_2$	4.07 ± 1.5	4.60 ± 1.4	9.80 ± 3.5	6.17 ± 3.0	14.47 ± 2.5	148.67 ± 18.6	104.50 ± 17.1
Prostate$_3$	2.53 ± 1.2	2.77 ± 1.4	11.27 ± 6.7	3.37 ± 1.1	3.53 ± 0.7	47.17 ± 8.8	38.43 ± 9.3
Prostate$_4$	2.77 ± 1.5	3.43 ± 1.8	7.30 ± 3.7	4.70 ± 3.0	19.73 ± 3.9	258.40 ± 82.9	191.77 ± 65.2
Average	2.88 ± 3.0	3.23 ± 3.1	10.16 ± 9.7	3.94 ± 2.9	21.59 ± 30.5	484.56 ± 495.7	369.81 ± 357.0

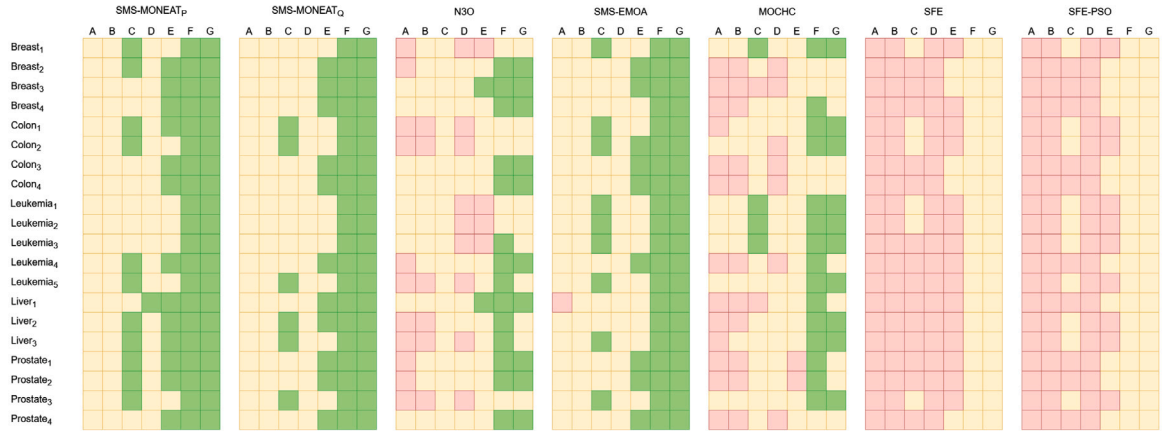


Fig. 3. Friedman's post hoc test to compare the number of features selected on each algorithm on the experiments performed on microarray datasets. Each column (A–G) represents the results for each algorithm against (A) SMS-MONEAT$_P$, (B) SMS-MONEAT$_Q$, (C) N3O, (D) SMS-EMOA, (E) MOCHC, (F) SFE and (G) SFE-PSO. Green and red squares represents that a significant difference on performance was found, positive and negative, respectively. Yellow squares represent that no significant difference was found between any two algorithms.

Table 9
Comparison using the Condorcet method regarding the number of selected features using the Friedman post hoc test results. The algorithm with the best performance according to this method is highlighted.

	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O	SMS-EMOA	MOCHC	SFE	SFE-PSO	Total
SMS-MONEAT$_P$	X	0	11	1	13	20	20	**65**
SMS-MONEAT$_Q$	0	X	6	0	11	20	20	**57**
N3O	0	0	X	0	2	14	10	**26**
SMS-EMOA	0	0	9	X	9	20	20	**58**
MOCHC	0	0	4	0	X	15	10	**29**
SFE	0	0	0	0	0	X	0	**0**
SFE-PSO	0	0	0	0	0	0	X	**0**

post hoc test results, a Condorcet contest was employed to compare the algorithms, shown in Table 9. The Condorcet contest indicates that SMS-MONEA$_P$ had the best performance overall. The algorithm performs significantly better at reducing the number of features on every 20/20 datasets against SFE and SFE-PSO, and on over half of them against N3O and MOCHC. It also showed better performance against SMS-EMOA in just one dataset, and no significant difference was found against SMS-MONEAT$_Q$.

6.2.2. *Binary cross-entropy*

The binary cross-entropy was employed to evaluate the classification models obtained within each algorithm. Table 10 shows the mean and standard deviation of the binary cross-entropy obtained from each

Table 10
Mean and standard deviation (±) of each experiment's binary cross-entropy (lower values are better ↓). The last row describes the average value from all of the experiments. The best mean value obtained by an algorithm in each dataset is highlighted.

Dataset	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O [8]	SMS-EMOA [59]	MOCHC [65]	SFE [32]	SFE-PSO [32]
Breast$_1$	0.1495 ± 0.18	0.1522 ± 0.15	0.1351 ± 0.16	8.6279 ± 19.56	7.0132 ± 17.80	0.0000 ± 0.00	0.0000 ± 0.00
Breast$_2$	0.5792 ± 0.25	0.5584 ± 0.18	0.4687 ± 0.12	16.3718 ± 10.05	13.9696 ± 9.44	13.3223 ± 8.15	14.8557 ± 9.29
Breast$_3$	0.4478 ± 0.09	0.4530 ± 0.09	0.4354 ± 0.12	17.7360 ± 6.57	14.0888 ± 5.09	16.2037 ± 5.87	15.6741 ± 5.51
Breast$_4$	0.6518 ± 0.17	0.6226 ± 0.18	0.6524 ± 0.23	27.2599 ± 12.91	22.9297 ± 10.25	25.7701 ± 10.02	24.0688 ± 9.89
Colon$_1$	0.2964 ± 0.44	0.3089 ± 0.44	0.2646 ± 0.30	7.6439 ± 13.22	7.4494 ± 10.85	4.3033 ± 10.04	4.3033 ± 10.04
Colon$_2$	0.1962 ± 0.25	0.2415 ± 0.37	0.1368 ± 0.12	8.8550 ± 15.76	9.6059 ± 15.81	0.0000 ± 0.00	1.7213 ± 6.55
Colon$_3$	0.1420 ± 0.07	0.1456 ± 0.07	0.1466 ± 0.07	2.2455 ± 3.57	1.2432 ± 2.30	1.2240 ± 2.63	1.2240 ± 2.63
Colon$_4$	0.4671 ± 0.25	0.4881 ± 0.30	0.4520 ± 0.28	16.4272 ± 12.91	12.1239 ± 11.67	21.7783 ± 12.42	20.2953 ± 12.73
Leukemia$_1$	0.3843 ± 0.85	0.3498 ± 0.62	0.4086 ± 0.84	12.3958 ± 22.21	10.6745 ± 21.03	6.0939 ± 16.14	5.1639 ± 15.76
Leukemia$_2$	0.3930 ± 0.91	0.3581 ± 0.88	0.3248 ± 0.46	12.9792 ± 22.36	8.8838 ± 19.46	6.8853 ± 17.85	7.7459 ± 18.13
Leukemia$_3$	0.1236 ± 0.29	0.0926 ± 0.13	0.0776 ± 0.08	3.2666 ± 8.27	0.9716 ± 4.71	0.0555 ± 0.30	0.0000 ± 0.00
Leukemia$_4$	0.2254 ± 0.11	0.2286 ± 0.11	0.1889 ± 0.13	3.7892 ± 6.47	3.2251 ± 5.73	1.1475 ± 2.98	1.7213 ± 3.50
Leukemia$_5$	0.3170 ± 0.19	0.3243 ± 0.18	0.3230 ± 0.22	9.7584 ± 11.34	8.1895 ± 13.09	4.4281 ± 8.92	8.7313 ± 13.49
Liver$_1$	0.3184 ± 0.10	0.3117 ± 0.09	0.2760 ± 0.11	3.5434 ± 2.92	2.3972 ± 2.37	2.2358 ± 2.10	2.3686 ± 2.11
Liver$_2$	0.3568 ± 0.41	0.4172 ± 0.50	0.4030 ± 0.41	13.9422 ± 22.20	16.4317 ± 23.01	13.7705 ± 22.21	12.9099 ± 21.17
Liver$_3$	0.1775 ± 0.12	0.1693 ± 0.12	0.1342 ± 0.09	2.4910 ± 3.76	1.1379 ± 2.61	0.8607 ± 2.23	0.6455 ± 1.97
Prostate$_1$	0.6851 ± 0.15	0.6683 ± 0.12	0.7095 ± 0.24	29.2038 ± 12.30	26.4111 ± 7.90	30.0080 ± 11.39	30.4652 ± 9.92
Prostate$_2$	0.7400 ± 0.20	0.6960 ± 0.15	0.7038 ± 0.24	31.1768 ± 10.80	25.9804 ± 10.17	25.3664 ± 11.18	26.1154 ± 12.44
Prostate$_3$	0.8654 ± 0.58	0.7307 ± 0.40	1.0628 ± 0.70	29.0371 ± 21.41	20.5691 ± 18.60	22.0421 ± 23.06	28.3439 ± 22.41
Prostate$_4$	0.6232 ± 0.18	0.6000 ± 0.15	0.5928 ± 0.17	15.0791 ± 10.60	12.8405 ± 9.35	9.6789 ± 6.76	9.6789 ± 6.32
Average	0.4070 ± 0.42	0.3958 ± 0.38	0.3948 ± 0.40	13.5915 ± 16.48	11.3068 ± 14.75	10.2587 ± 14.68	10.8016 ± 14.94

Table 11
Comparison using the Condorcet method regarding the binary cross-entropy using the Friedman post hoc test results. The algorithm with the best performance according to this method is highlighted.

	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O	SMS-EMOA	MOCHC	SFE	SFE-PSO	Total
SMS-MONEAT$_P$	X	0	0	11	9	13	15	**48**
SMS-MONEAT$_Q$	0	X	0	11	10	14	16	**51**
N3O	0	0	X	11	9	13	13	**46**
SMS-EMOA	0	0	0	X	0	0	0	**0**
MOCHC	0	0	0	0	X	0	0	**0**
SFE	3	3	3	0	0	X	0	**9**
SFE-PSO	2	2	2	0	0	0	X	**6**

dataset and algorithm. N3O obtained the best mean value regarding this indicator. However, SMS-MONEAT$_P$ and SMS-MONEAT$_Q$ have a competitive performance respect to N3O. On the other hand, SMS-EMOA obtained the worst average over the loss function.

The Friedman post hoc test was also computed for the loss function, and the results are shown in Fig. 4. Moreover, the Condorcet contest was also used to compare the algorithms' performance and the results are shown in Table 11. Overall, SMS-MONEAT$_P$, SMS-MONEAT$_Q$ and N3O showed the best performance on the loss function indicated by the Condorcet contest, with SMS-MONEAT$_Q$ the one with the highest value. A relevant insight from this indicator is that there were no significant differences between SMS-MONEAT and N3O results. Nevertheless, as mentioned in Section 6.2.1, SMS-MONEAT$_P$ significantly reduced features over N3O on 11/20 datasets (shown in Table 9). This means that SMS-MONEAT can significantly reduce the number of features without reducing the classification performance. Following the same idea, SMS-MONEAT solutions showed significantly better performance on the loss function on 11/20 datasets over SMS-EMOA (shown in Table 11), showing the advantages of integrating the neuroevolution part of SMS-MONEAT over the conventional codification of SMS-EMOA.

Finally, even though SFE algorithms use only the binary cross entropy as optimisation function, SMS-EMOA algorithms showed a significantly better performance on this indicator in close to a third of the datasets.

6.2.3. Geometric mean

The geometric mean was employed to evaluate the classification performance of the models obtained by each algorithm. This metric was selected because it considers each class separately, which is ideal for imbalanced datasets. In contrast with the previous indicators aimed to be minimised, the geometric mean should be maximised. Table 12 shows the mean and standard deviation (±) obtained for each experiment. SFE showed, on average, the highest geometric mean, followed by SFE-PSO. This may reflect better generalisation; however, it could also be related to the larger number of features selected by these algorithms compared to the rest. On the other hand, the fact that SMS-MONEAT$_P$, SMS-MONEAT$_Q$, and N3O achieved a lower average on this indicator while obtaining a better average on the binary cross-entropy (shown in Table 10) might indicate that the classification models obtained by these algorithms are overfitting to the training datasets.

Similar to the previous metrics, a Friedman post hoc test was employed to investigate statistical differences between each algorithm regarding the geometric mean results, shown on Fig. 5. Moreover, a Condorcet contest was performed using the statistical differences found within the post hoc test and the results are shown in Table 13. SFE and SFE-PSO obtained the highest values on the Condorcet contest. Both algorithms performed significantly better on three datasets against SMS-MONEAT$_P$: Leukemia$_4$, Liver$_1$ and Prostate$_4$. As mentioned previously, this could be related to the number of features that SFE and SFE-PSO selected, which is significantly larger than SMS-MONEAT's results for these 3 datasets or that SMS-MONEAT classification models might be overfitting to the loss function. Nevertheless, since SMS-MONEAT is a multiobjective optimisation method, more metrics, such as the geometric mean, could be included in future work to prevent this.

6.2.4. Hypervolume

The hypervolume of the final population in each run, calculated using the number of features selected and the binary cross-entropy, was used as another indicator to compare the algorithms' performance. A higher hypervolume value indicates a closer approximation to the

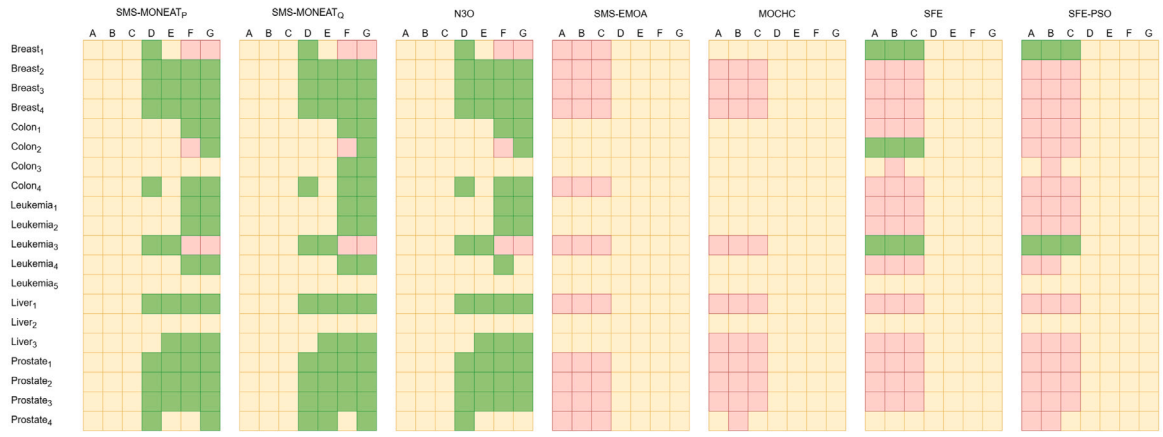


Fig. 4. Friedman's post hoc test to compare the binary cross-entropy obtained on each algorithm on the experiments performed on microarray datasets. Each column represents the results for each algorithm against (A) SMS-MONEAT$_P$, (B) SMS-MONEAT$_Q$, (C) N3O, (D) SMS-EMOA, (E) MOCHC, (F) SFE and (G) SFE-PSO. Green and red squares represents that a significant difference on performance was found, positive and negative, respectively. Yellow squares represent that no significant difference was found between both algorithms.

Table 12
Mean and standard deviation (±) of each experiment's geometric mean (higher values are better ↑). The last row describes the average value from all of the experiments. The best mean value obtained by an algorithm in each dataset is highlighted.

Dataset	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O [8]	SMS-EMOA [59]	MOCHC [65]	SFE [32]	SFE-PSO [32]
Breast$_1$	0.9197 ± 0.25	0.9867 ± 0.02	0.9573 ± 0.18	0.8320 ± 0.38	0.8588 ± 0.34	1.0000 ± 0.00	1.0000 ± 0.00
Breast$_2$	0.7424 ± 0.14	0.7562 ± 0.14	0.7879 ± 0.11	0.7496 ± 0.14	0.7894 ± 0.14	0.8151 ± 0.11	0.7960 ± 0.12
Breast$_3$	0.8052 ± 0.09	0.7945 ± 0.07	0.8088 ± 0.10	0.7807 ± 0.09	0.8309 ± 0.06	0.8038 ± 0.07	0.8119 ± 0.06
Breast$_4$	0.6329 ± 0.17	0.6451 ± 0.16	0.6344 ± 0.15	0.6024 ± 0.18	0.6207 ± 0.14	0.6131 ± 0.18	0.6257 ± 0.19
Colon$_1$	0.9161 ± 0.12	0.9247 ± 0.13	0.9025 ± 0.21	0.8970 ± 0.20	0.9052 ± 0.14	0.9505 ± 0.12	0.9505 ± 0.12
Colon$_2$	0.8971 ± 0.27	0.8874 ± 0.27	0.9512 ± 0.11	0.7776 ± 0.38	0.8552 ± 0.27	1.0000 ± 0.00	0.9805 ± 0.07
Colon$_3$	0.9624 ± 0.04	0.9678 ± 0.04	0.9714 ± 0.03	0.9703 ± 0.04	0.9876 ± 0.02	0.9877 ± 0.03	0.9877 ± 0.03
Colon$_4$	0.7963 ± 0.15	0.8174 ± 0.14	0.8355 ± 0.14	0.7655 ± 0.21	0.8252 ± 0.15	0.7128 ± 0.20	0.7201 ± 0.20
Leukemia$_1$	0.8768 ± 0.30	0.8626 ± 0.30	0.8435 ± 0.34	0.7480 ± 0.42	0.7813 ± 0.40	0.8858 ± 0.31	0.9000 ± 0.31
Leukemia$_2$	0.8622 ± 0.34	0.8875 ± 0.30	0.8102 ± 0.37	0.7569 ± 0.43	0.8155 ± 0.37	0.8667 ± 0.35	0.8569 ± 0.35
Leukemia$_3$	0.9673 ± 0.08	0.9708 ± 0.08	0.9734 ± 0.08	0.9576 ± 0.09	0.9832 ± 0.06	0.9965 ± 0.02	1.0000 ± 0.00
Leukemia$_4$	0.8903 ± 0.11	0.9153 ± 0.10	0.9379 ± 0.06	0.9432 ± 0.09	0.9627 ± 0.06	0.9884 ± 0.03	0.9826 ± 0.04
Leukemia$_5$	0.8276 ± 0.22	0.8394 ± 0.20	0.8309 ± 0.15	0.8779 ± 0.13	0.8709 ± 0.20	0.9444 ± 0.10	0.8818 ± 0.21
Liver$_1$	0.8999 ± 0.06	0.9004 ± 0.07	0.9248 ± 0.05	0.9450 ± 0.03	0.9483 ± 0.03	0.9540 ± 0.03	0.9510 ± 0.04
Liver$_2$	0.8022 ± 0.33	0.7885 ± 0.36	0.7201 ± 0.41	0.7389 ± 0.42	0.7016 ± 0.44	0.7471 ± 0.43	0.7707 ± 0.40
Liver$_3$	0.9449 ± 0.06	0.9520 ± 0.06	0.9823 ± 0.04	0.9138 ± 0.25	0.9581 ± 0.18	0.9914 ± 0.02	0.9935 ± 0.02
Prostate$_1$	0.5975 ± 0.15	0.5958 ± 0.13	0.6012 ± 0.17	0.5431 ± 0.23	0.5946 ± 0.12	0.5661 ± 0.17	0.5570 ± 0.15
Prostate$_2$	0.6019 ± 0.14	0.5979 ± 0.16	0.6015 ± 0.14	0.5315 ± 0.22	0.5992 ± 0.14	0.6226 ± 0.18	0.6007 ± 0.19
Prostate$_3$	0.3886 ± 0.43	0.5300 ± 0.40	0.3593 ± 0.40	0.3719 ± 0.42	0.4857 ± 0.42	0.5552 ± 0.45	0.4259 ± 0.42
Prostate$_4$	0.6240 ± 0.15	0.6352 ± 0.18	0.6540 ± 0.18	0.7992 ± 0.13	0.8210 ± 0.12	0.8498 ± 0.09	0.8502 ± 0.08
Average	0.7978 ± 0.26	0.8128 ± 0.24	0.8044 ± 0.26	0.7751 ± 0.30	0.8098 ± 0.27	0.8425 ± 0.25	0.8321 ± 0.25

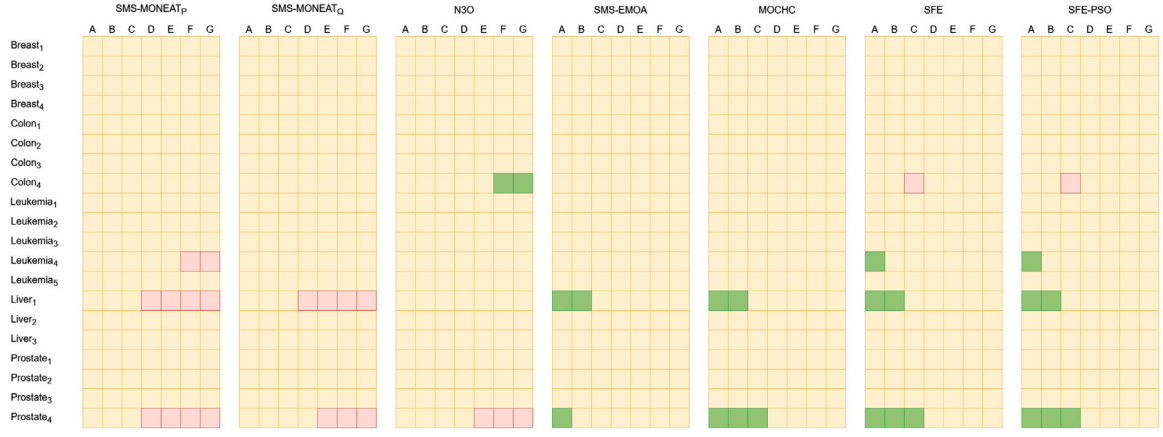


Fig. 5. Friedman's post hoc test to compare the geometric mean obtained on each algorithm on the experiments performed on microarray datasets. Each column represents the results for each algorithm against (A) SMS-MONEAT$_P$, (B) SMS-MONEAT$_Q$, (C) N3O, (D) SMS-EMOA, (E) MOCHC, (F) SFE and (G) SFE-PSO. Green and red squares represents that a significant difference on performance was found, positive and negative, respectively. Yellow squares represent that no significant difference was found between both algorithms.

Table 13
Comparison using the Condorcet method regarding the geometric mean using the Friedman post hoc test results. The algorithm with the best performance according to this method is highlighted.

	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O	SMS-EMOA	MOCHC	SFE	SFE-PSO	Total
SMS-MONEAT$_P$	X	0	0	0	0	0	0	**0**
SMS-MONEAT$_Q$	0	X	0	0	0	0	0	**0**
N3O	0	0	X	0	0	1	1	**2**
SMS-EMOA	2	1	0	X	0	0	0	**3**
MOCHC	2	2	1	0	X	0	0	**5**
SFE	3	2	1	0	0	X	0	**6**
SFE-PSO	3	2	1	0	0	0	X	**6**

Table 14
Mean and standard deviation (±) of each experiment's population hypervolume (higher values are better ↑). The last row describes the average value from all of the experiments. The best mean value obtained by an algorithm in each dataset is highlighted.

Dataset	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O [8]	SMS-EMOA [59]	MOCHC [65]
Breast$_1$	892.27 ± 2.6	892.91 ± 1.2	830.24 ± 78.0	841.32 ± 194.9	890.84 ± 2.6
Breast$_2$	886.46 ± 2.6	888.24 ± 1.3	829.69 ± 26.6	761.68 ± 105.2	762.55 ± 54.4
Breast$_3$	887.45 ± 1.4	888.01 ± 1.2	855.14 ± 19.9	687.72 ± 87.6	363.36 ± 37.8
Breast$_4$	885.46 ± 2.5	886.96 ± 1.5	837.81 ± 23.7	627.16 ± 148.1	803.58 ± 88.6
Colon$_1$	890.14 ± 5.9	892.23 ± 1.9	825.54 ± 72.4	824.80 ± 140.6	864.91 ± 62.3
Colon$_2$	891.36 ± 3.2	893.30 ± 0.6	819.86 ± 78.3	787.62 ± 202.7	885.78 ± 5.8
Colon$_3$	891.95 ± 1.0	892.07 ± 0.9	851.40 ± 63.7	870.68 ± 41.3	697.51 ± 104.7
Colon$_4$	887.64 ± 3.3	889.22 ± 1.9	842.24 ± 23.6	720.60 ± 167.3	702.35 ± 110.4
Leukemia$_1$	889.54 ± 10.5	892.53 ± 2.3	846.33 ± 39.6	783.83 ± 266.5	814.12 ± 233.9
Leukemia$_2$	889.95 ± 11.4	892.77 ± 2.0	847.82 ± 22.5	751.65 ± 296.6	892.97 ± 2.1
Leukemia$_3$	892.53 ± 3.6	893.34 ± 0.7	844.15 ± 64.4	870.73 ± 82.8	892.03 ± 2.5
Leukemia$_4$	890.88 ± 1.6	891.66 ± 1.1	852.91 ± 26.8	869.35 ± 48.2	849.84 ± 10.4
Leukemia$_5$	890.33 ± 2.5	891.00 ± 1.8	841.21 ± 26.3	823.14 ± 140.7	866.35 ± 97.1
Liver$_1$	889.33 ± 1.5	889.83 ± 1.1	860.85 ± 25.6	862.03 ± 28.8	527.49 ± 23.8
Liver$_2$	890.73 ± 3.9	892.30 ± 1.3	846.51 ± 21.6	736.06 ± 294.2	878.42 ± 7.0
Liver$_3$	891.49 ± 1.7	892.42 ± 1.0	848.20 ± 33.2	859.72 ± 55.1	873.21 ± 9.5
Prostate$_1$	885.07 ± 1.6	886.11 ± 1.4	829.88 ± 21.0	605.44 ± 104.2	775.16 ± 62.8
Prostate$_2$	884.67 ± 2.0	886.12 ± 1.0	834.71 ± 17.7	555.59 ± 115.9	735.64 ± 83.8
Prostate$_3$	883.74 ± 8.3	890.63 ± 2.3	823.04 ± 38.4	690.62 ± 268.5	762.29 ± 194.9
Prostate$_4$	885.33 ± 2.3	886.10 ± 1.8	850.30 ± 19.9	760.77 ± 111.1	846.70 ± 50.0
Average	888.82 ± 5.4	890.39 ± 2.9	840.89 ± 43.4	764.53 ± 188.0	784.26 ± 157.6

optimal solution and greater diversity in the population. As with the previous indicators, Table 14 presents the mean and standard deviation (±) of the hypervolume values for each dataset and algorithm. The algorithm with the best mean value for each dataset is highlighted in green in Table 14. As seen, SMS-MONEAT$_Q$ achieve the highest average and has the higher value in 19 out of 20 datasets. The fact that SMS-MONEAT$_Q$ achieve a higher hypervolume value over SMS-MONEAT$_P$ indicated the advantage of including the external archive with the speciation method to increase the diversity of the population.

For this analysis, the SFE and SFE-PSO algorithms were excluded because they selected significantly more features than the other algorithms. This would require setting a disproportionately large reference point to compute the hypervolume, which in turn, would diminish the observable differences between the remaining algorithms.

The Friedman post hoc test was computed to investigate the statistical differences between the algorithm's performance over their populations' hypervolume, and the results are shown in Fig. 6. Furthermore, a Condorcet contest was employed to analyse these statistical differences, shown in Table 15. SMS-MONEAT$_Q$ obtained the highest value in the Condorcet contest, followed by SMS-MONEAT$_P$. These results show that the populations obtained from SMS-MONEAT approximate closer to the real Pareto front and have a higher diversity of solutions than the rest of the algorithms. This is expected compared to N3O since SMS-MONEAT includes the number of features as an optimisation function. On the other hand, comparing the multiobjective optimisation approaches, SMS-MONEAT, SMS-EMOA, and MOCHC, the benefits of using a neuroevolution approach are shown with a higher hypervolume value.

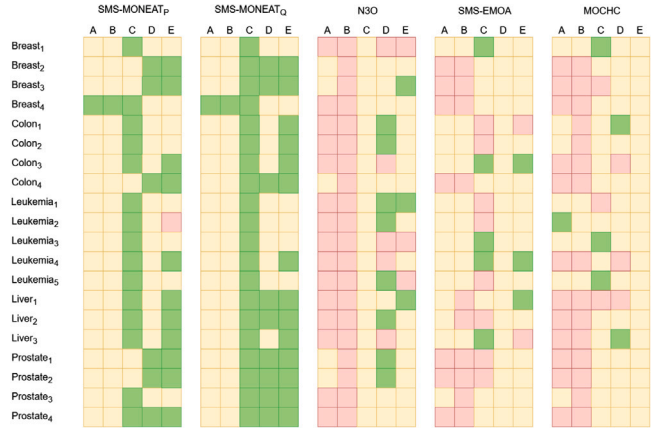


Fig. 6. Friedman's post hoc test to compare the population's hypervolume obtained on each algorithm on the experiments performed on microarray datasets. Each column represents the results for each algorithm against (A) SMS-MONEAT$_P$, (B) SMS-MONEAT$_Q$, (C) N3O, (D) SMS-EMOA, and (E) MOCHC. Green and red squares represents that a significant difference on performance was found, positive and negative, respectively. Yellow squares represent that no significant difference was found between any two algorithms.

6.2.5. Execution time

Execution time was the last metric employed for comparison. As mentioned, all the algorithms execute the same number of fitness evaluations (18 000); however, the evaluations differ from the neuroevolution-based approaches from the rest of the approaches. For SMS-MONEAT and N3O, which are neuroevolution-based algorithms,

Table 15
Comparison using the Condorcet method regarding the hypervolume using the Friedman post hoc test results. The algorithm with the best performance according to this method is highlighted.

	SMS-MONEAT$_P$	SMS-MONEAT$_Q$	N3O	SMS-EMOA	MOCHC	Total
SMS-MONEAT$_P$	X	0	15	7	12	**34**
SMS-MONEAT$_Q$	0	X	20	10	15	**45**
N3O	0	0	X	8	3	**11**
SMS-EMOA	0	0	5	X	3	**8**
MOCHC	1	0	4	2	X	**7**

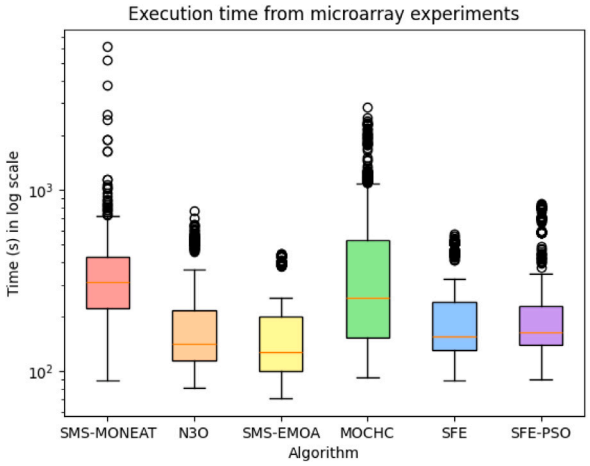


Fig. 7. Execution time for each algorithm run on the microarray datasets.

an evaluation means building an ANN from the genotype and evaluating it in the training dataset. On the other hand, for SMS-EMOA, MOCHC, SFE and SFE-PSO, an evaluation implies training a K-means model with a subset of the training dataset and then evaluating the model in the rest of the dataset. For simplicity, we assumed those fitness evaluations were equivalent.

Fig. 7 shows the boxplot of the execution time of all the independent runs for each algorithm. SMS-EMOA has the lowest mean value (174.89 s), followed by N3O (205.64 s), SFE (209.95 s), SFE-PSO (235.13 s), SMS-MONEAT (386.12 s) and finally MOCHC (460.83 s). It was expected that SMS-MONEAT and MOCHC requires more time than the mono-objective approaches. However, the reason that SMS-EMOA took the lowest time might be due to two reasons: (i) using a binary codification might require less time than using the crossover operator from the neuroevolution approaches, and (ii) the population of SMS-EMOA is initialised with fewer features selected than SFE, SFE-PSO and MOCHC and therefore, training the KNN model might require less time.

For this metric, SMS-MONEAT was not split between SMS-MONEAT$_P$ and SMS-MONEAT$_Q$ since they were executed together; nevertheless, excluding the external archive could reduce SMS-MONEAT$_P$ execution time. Although SMS-MONEAT's execution time is slower than most of the other algorithms, it remains within a reasonable range for practical applications. The trade-off between speed and the additional benefits provided by the proposed approach justifies its use in microarray classification.

6.2.6. Computational complexity

The computational complexity of SMS-MONEAT depends on various variables: first, the number of iterations (I), the population size (P), the fitness function cost ($f(x)$), and the number of objectives. Also, during its execution, the number of neurons and connections (C) from the individuals in the population will increase, which will also increase the complexity of the crossover operator.

Another important factor to consider when using SMS-MONEAT is the number of objectives since it computes the population hypervolume. For 2 or 3 objectives, the best-known algorithms have a time complexity of $O(n \log n)$; nevertheless, for a higher number of objectives, the complexity becomes $O(n^{k/2})$, making it computationally challenging as the number of objectives increases [73].

The computational complexity of SMS-MONEAT for two optimisation objectives would be as $O(I \cdot (f(x) + C_i^2 + 2P^2 + PlogP + \delta))$. The reason for this is that in each iteration, the fitness function, a crossover operator (whose complexity changes in each iteration depending on the number of connections of the parent individuals, but also not in every iteration the crossover is computed) and fast non-dominated sorting algorithm and the hypervolume of the population are computed. The δ value includes constant values such as the mutation operators.

Similarly, the computational complexity of N3O will then be $O(I \cdot P \cdot (f(x) + C_i^2 + \delta))$. The difference from SMS-MONEAT is that it does not include the multiobjective parts, such as the fast non-dominated sorting and the hypervolume computations.

On the other hand, the computational complexity of SMS-EMOA for two optimisation objectives would be as $O(I \cdot (f(x) + 2P^2 + PlogP + \delta))$. The main difference from SMS-MONEAT is that the crossover and mutation operators in SMS-EMOA have a constant complexity, and they require less time than the operators in SMS-MONEAT.

The computational complexity of MOCHC should be similar to the one from SMS-EMOA, since the main difference would be that SMS-EMOA computes the hypervolume of the population while MOCHC uses the crowding distance. Both methods have a complexity of $O(PlogP)$ for two objectives.

Finally, the computational complexity of SFE would be $O(I \cdot f(x))$ since the mutation operator takes constant time and only has an individual in the population. For SFE-PSO, it would depend on how many iterations are employed for the first part where the SFE operators are used and how many iterations are employed for the PSO algorithm. The computational complexity of PSO is $O(I \cdot P \cdot (f(x) + D))$, where D is the number of features since in each iteration, the particles should be updated.

6.3. Benchmarking against a SOTA hybrid feature selection algorithm

A critical and insightful further comparison was conducted with a state-of-the-art approach recently presented by Xie et al. [48]. Their innovative work introduced a hybrid multi-objective feature selection method, skillfully integrating an improved Max-Relevance and Min-Redundancy (mRMR) filter with an enhanced multilayer binary firefly algorithm (MBFA) operating as a wrapper. The complexity of their evaluation was amplified by the sophisticated fitness function they employed: a stacking ensemble learning classifier underpinned by an attention mechanism, which is a state-of-the-art AI component. This required the training of four distinct classification models – specifically Random Forest, Parsimonious Bayes, K-Nearest Neighbour, and Adaboost – with each model undergoing a rigorous 5-fold cross-validation process. Consequently, a substantial computational cost was incurred, requiring the training of 20 individual classification models for every single evaluation within their optimisation process.

While the precise number of evaluations was not explicitly stated in their publication, we meticulously estimated it based on the provided parameters using the formula: $E = P \times R \times (I + 1) \times M$, where E represents the total number of evaluations, P the population size, R the number of MBFA runs, I the number of iterations (with one added to

Table 16
Comparison between the proposed method SMS-MONEAT and a state-of-the-art method mRMR-MBFA on multiple binary classification microarray datasets according to the accuracy and the number of selected features (SF).

Dataset	Samples	Features	Method	Accuracy (%)	SF
Colon [74]	62	2000	mRMR-MBFA [48] (mean)	90.48	9.4
			SMS-MONEAT$_P$ (mean)	77.54	2.9
			SMS-MONEAT$_P$ (best)	100	2
			SMS-MONEAT$_Q$ (mean)	80.32	3.5
			SMS-MONEAT$_Q$ (best)	100	2
DLBCL [75]	77	7070	mRMR-MBFA [48] (mean)	98.75	5.9
			SMS-MONEAT$_P$ (mean)	88.21	4.1
			SMS-MONEAT$_P$ (best)	100	1
			SMS-MONEAT$_Q$ (mean)	89.52	3.9
			SMS-MONEAT$_Q$ (best)	100	2
Leukemia$_5$ [68]	72	7129	mRMR-MBFA [48] (mean)	100	4.3
			SMS-MONEAT$_P$ (mean)	89.88	4.4
			SMS-MONEAT$_P$ (best)	100	2
			SMS-MONEAT$_Q$ (mean)	90.30	4.6
			SMS-MONEAT$_Q$ (best)	100	2
Prostate$_4$ [70]	136	12 600	mRMR-MBFA[48] (mean)	94.18	6.7
			SMS-MONEAT$_P$ (mean)	69.56	4.1
			SMS-MONEAT$_P$ (best)	100	4
			SMS-MONEAT$_Q$ (mean)	72.51	4.5
			SMS-MONEAT$_Q$ (best)	100	4

account for the initial population), and M the number of models trained per evaluation. Based on their reported parameters – a population size of 100 individuals, MBFA executed 9 times across all layers, with 3 iterations each – the total number of evaluations undertaken in their study was calculated to be a significant 72,000.

To ensure a comparison as fair as possible, SMS-MONEAT was also run for 72,000 evaluations, matching the number used by the mRMR-MBFA method. It is important to note, however, that the hyperparameter optimisation for SMS-MONEAT, facilitated by the efficient iRace algorithm (detailed in Section 5.2), was conducted with a considerably lower computational cost of only 18,000 evaluations. Furthermore, to mitigate the risk of premature convergence, we implemented a robust restart mechanism within SMS-MONEAT. This mechanism strategically reinitialised the population whenever the best observed value of the loss function failed to improve over a sustained period of 3000 consecutive iterations, demonstrating a proactive approach to maintaining exploration and solution diversity throughout the optimisation process.

Consistent with our preceding experimental setup, we employed a rigorous stratified 10-fold cross-validation with three repetitions to thoroughly evaluate the performance of SMS-MONEAT on four distinct binary classification microarray datasets. The results of this direct comparison between SMS-MONEAT and the mRMR-MBFA method, detailing both the classification accuracy and the number of features selected, are comprehensively presented in Table 16.

As shown in Table 16, the proposed SMS-MONEAT method does not surpass the mean accuracy reported for the mRMR-MBFA approach. However, the best-performing solutions obtained by SMS-MONEAT on each dataset highlight its competitive potential, achieving 100% accuracy while using fewer features. This reduction in the number of features is particularly valuable for the task at hand, as it enhances the interpretability of the results.

As previously highlighted, a fundamental and noteworthy distinction between our proposed SMS-MONEAT and other methodologies, such as the mRMR-MBFA approach, lies in the execution of the prediction step. SMS-MONEAT directly leverages the classification model for prediction, whereas mRMR-MBFA necessitates a preceding model training phase before predictions can be made. This inherent methodological divergence underscores one of the significant challenges encountered when striving for equitable comparisons across different feature selection paradigms.

6.4. Concluding remarks on the comparative evaluation

As highlighted in Section 2.1, the landscape of feature selection methodologies for microarray datasets is rich and diverse. Nevertheless, a direct and meaningful comparison of published results across these various approaches presents significant inherent challenges. In this section, we aim to articulate the critical issues we encountered during our rigorous efforts to establish a comprehensive comparative framework.

A primary obstacle lies in the inconsistent used of evaluation metrics. While a majority of prior studies predominantly relied on accuracy to assess the performance of classification models, this metric suffers from a critical limitation when applied to the often imbalanced class distributions prevalent in microarray datasets. Recognising this deficiency, our study adopted the geometric mean as a more robust and representative performance indicator.

Furthermore, the crucial aspect of computational effort is frequently overlooked in the reporting of results. The number of fitness function evaluations serves as a cornerstone for enabling fair and objective comparisons between optimisation algorithms. Regrettably, a substantial number of studies present findings on microarray datasets without explicitly considering or reporting this vital parameter, hindering a true understanding of the computational resources invested.

The heterogeneity of fitness functions across different approaches further complicates direct comparisons. The design of the fitness function is intrinsically linked to the chosen classification method. For instance, our methodology uniquely employs a fitness function that involves constructing and evaluating an ANN directly from the encoded genotype. In stark contrast, a multitude of other works necessitate the computationally intensive training of multiple distinct classification models within their evaluation loop.

Therefore, in this work, we combined two different approaches to compare our proposed methodology. The first one, where we could perform our experiments with five algorithms and set them to equivalent hyperparameter optimisation and a control set up (Section 6.2), and the second one, where we just employed the results reported in the literature of sixth methodology (Section 6.3). This deliberately broad and meticulously executed comparative framework, encompassing both established and cutting-edge algorithms spanning the domains of multi-objective optimisation and feature selection, provides a substantially more robust, nuanced, and ultimately convincing validation of the merits of our proposed SMS-MONEAT approach. By rigorously benchmarking against a diverse set of competitive methodologies and employing a comprehensive set of evaluation metrics, we effectively

demonstrate the clear and compelling strengths and advantages of our work from multiple critical perspectives, thereby significantly amplifying the overall impact, credibility, and contribution of this paper to the field.

7. Conclusions and future work

This work introduced SMS-MONEAT, a neuroevolution-based multiobjective algorithm which is characterised by (i) handling multiple conflicting optimisation objectives by employing SMS-EMOA framework, (ii) optimising the topology and weights of artificial neural networks by using NEAT genetic encoding, and (iii) performing feature selection by utilising N3O evolutionary operators. Furthermore, a variation of SMS-MONEAT was introduced, which included a speciation method to maintain the diversity of subsets of features selected within an external archive.

The algorithms underwent testing in a feature selection and microarray classification methodology. They were compared against traditional and state-of-the-art wrapper methods for feature selection on microarray datasets.

When comparing the neuroevolution approaches, N3O and SMS-MONEAT, the results showed that SMS-MONEAT significantly reduced the number of features compared to N3O while generating classification models with similar performance. On the other hand, when comparing SMS-MONEAT and SMS-EMOA, the former achieved better values on the loss function and geometric mean, showing the advantages of integrating the neuroevolution bit.

SMS-MONEAT showed a promising performance and overcame most of the algorithms either on the number of features or the binary cross-entropy, which were its optimisation functions. Furthermore, it showed a competitive performance on the geometric mean indicator, it showed a significantly lower performance in three datasets against SFE, but this might be related to the high number of features selected by SFE which allows this algorithm a better generalisation.

Additionally, SMS-MONEAT was shown to be slower than the most of other methods. However, the trade-off between speed and the additional benefits SMS-MONEAT justifies its use in practical applications.

Finally, the advantages of including the speciation method in SMS-MONEAT were mainly observed with the results from the hypervolume were the external archive got a higher value indicating more diversity. Furthermore, the solutions from the external archive showed a higher value on the geometric mean which might indicate to achieve better generalisation.

Even though the experiments within this work were conducted on microarray datasets, SMS-MONEAT is a general method for feature selection and optimising ANNs on multiple objectives. These ANNs have a feed-forward architecture and can be employed in different tasks, such as regression and classification, as well as reinforcement learning tasks. Furthermore, SMS-MONEAT is a multiobjective optimisation algorithm scalable to more objectives than the two tested in this work. A further analysis of SMS-MONEAT's performance has to be done in the future. For example, we can test its performance with more optimisation objectives or in different tasks such as microarray multiclass classification. Additionally, it can be tested in a context different from microarray classification. Furthermore, another test that should be carried out would be to compare against manually design deep learning architectures within the SMS-EMOA framework, to observe the benefits of the neuroevolution feature.

For future work, we could modify the filter method employed in this work for an ensemble of filter methods. Some of the approaches presented in Table 1 have employed this and obtained promising results [15,20,24,36,45].

Additionally, the SMS-MONEAT design could be modified. For example, we suggest changing the hypervolume in the reduction step for different quality indicators, such as R2, and comparing their performance. On the other hand, other NEAT variations could be adapted within the SMS-EMOA framework, such as HyperNEAT [76]. Moreover, other genetic encodings and evolutionary operators could be compared. For example, a new algorithm could be designed using the Spectrum-diverse Unified Neuroevolution architecture (SUNA) within the SMS-EMOA framework to develop the SMS-MOSUNA algorithm [77].

Finally, another direction for future work, a training step using a gradient descent method could be integrated during evaluation, reducing the burden on the evolutionary algorithm to optimise ANN weights and allowing it to focus on evolving the network architecture. Although this would increase the computational cost of the evaluation function, it could represent a worthwhile trade-off between computational effort and improved classification performance.

CRediT authorship contribution statement

Daniel García-Núñez: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Katya Rodrígez-Vázquez:** Writing – review & editing, Supervision, Project administration, Methodology. **Carlos Hernández:** Writing – review & editing, Supervision, Project administration, Methodology. **Edgar Galván:** Writing – review & editing, Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this work is from the Curated Microarray Database which is open-access and the information can be find within the manuscript.

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